

ARAP · AIRNAV RISK ANALYSIS PLATFORM

# User Guide

How to plan, run, review and approve safety risk assessments  
with the twelve methods of the AirNav Risk Analysis Manual

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Software version 0.5 · Guide edition 5 · September 2026

# Contents

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1. Getting started
2. The dashboard
3. Projects and assessments
4. Working in a method study
5. Worksheet methods
6. Bowtie analysis
7. LOPA
8. Fault tree analysis
9. Fatigue modelling
10. Bayesian belief networks
11. STPA
12. FRAM
13. Extended methods (Manual chapters 18–31)
14. The hazard log
15. Actions
16. Review, approval and locking
17. Reports and exports
18. Administration
19. Troubleshooting and FAQ
20. What version 0.5 does not do yet

## ARAP User Guide

### AirNav Risk Analysis Platform — version 0.5

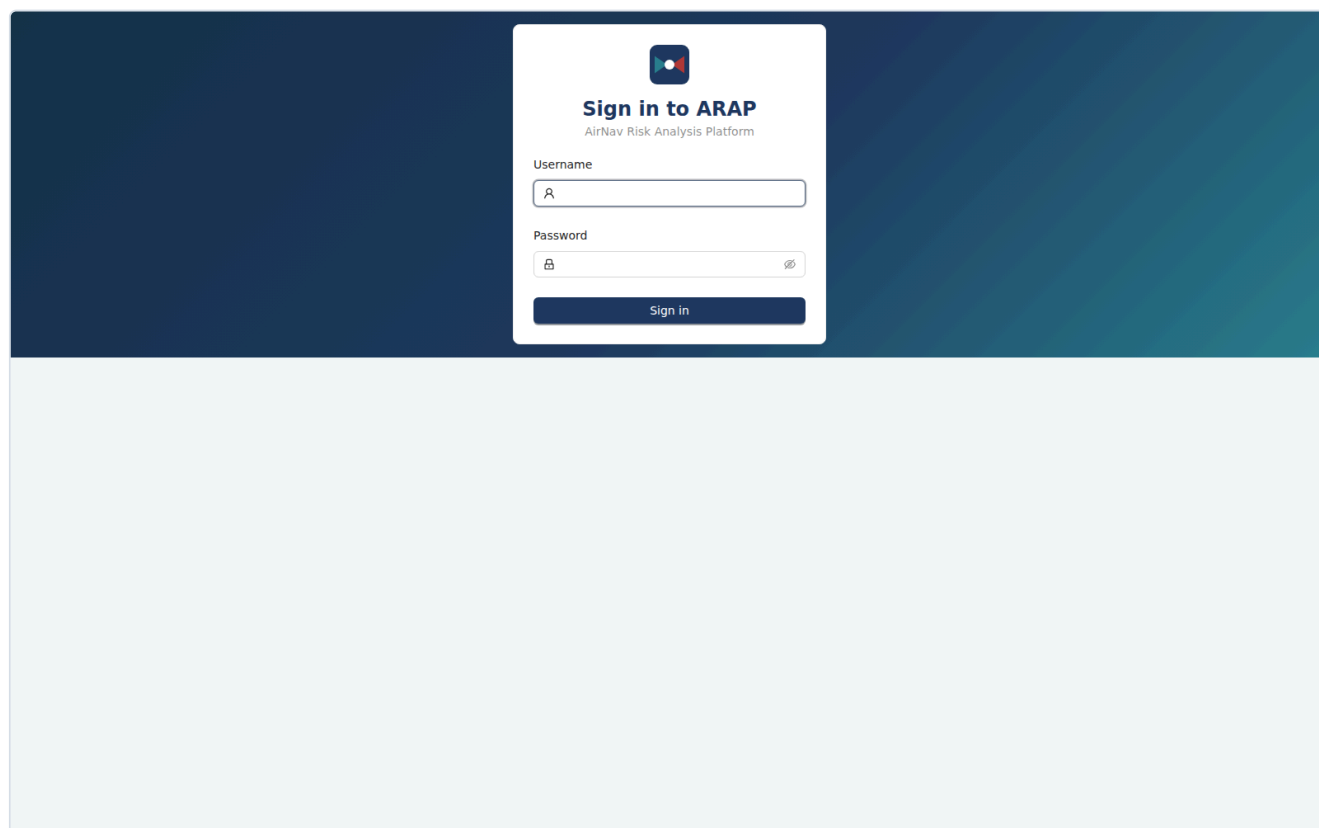
ARAP is the web application that implements the *AirNav Risk Analysis Manual*. It lets AirNav staff plan, run, review and approve safety risk assessments with twenty-six methods, keeps every hazard in one hazard log, and produces the Safety Assessment Report.

This guide explains how to use the application. For *why* and *when* to use each method, see the Manual ( [docs/AirNav\\_Risk\\_Analysis\\_Manual.pdf](#) ). For the formal requirements, see the SRS ( [docs/AirNav\\_Risk\\_Analysis\\_Software\\_Requirements.pdf](#) ).

# 1. Getting started

## 1.1 Signing in

Open ARAP in a browser (Chrome, Edge, Firefox or Safari) at the address your administrator gives you — for a local installation this is `http://localhost:8080` . Enter your username and password.



Sign-in page

## 1.2 Roles

What you can do depends on your role:

| Role             | Can do                                                                                                                                                       |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Viewer</b>    | Read everything in their scope; update actions they own.                                                                                                     |
| <b>Assessor</b>  | Create projects, assessments and method studies; edit worksheets and diagrams; send hazards to the hazard log; raise actions; submit assessments for review. |
| <b>Reviewer</b>  | Everything an assessor can, plus return, endorse or reject assessments, and read the audit trail.                                                            |
| <b>Authority</b> | Accept (or reject) residual risk, but only for the risk regions assigned to them (for example <i>Tolerable (lower)</i> ).                                    |
| <b>Admin</b>     | Manage users, roles and the risk classification scheme. Admin is not a substitute for the proper acceptance authority in normal use.                         |

## 1.3 Demo accounts

When ARAP is installed with demo data ( `ARAP_SEED_DEMO=true` ), these accounts exist, all with password `demo1234` :

| Username | Role                              | May accept                                       |
|----------|-----------------------------------|--------------------------------------------------|
| assessor | Assessor                          | —                                                |
| reviewer | Reviewer                          | —                                                |
| director | Authority                         | Tolerable (lower), Acceptable                    |
| accexec  | Authority (Accountable Executive) | Tolerable (upper), Tolerable (lower), Acceptable |
| viewer   | Viewer                            | —                                                |

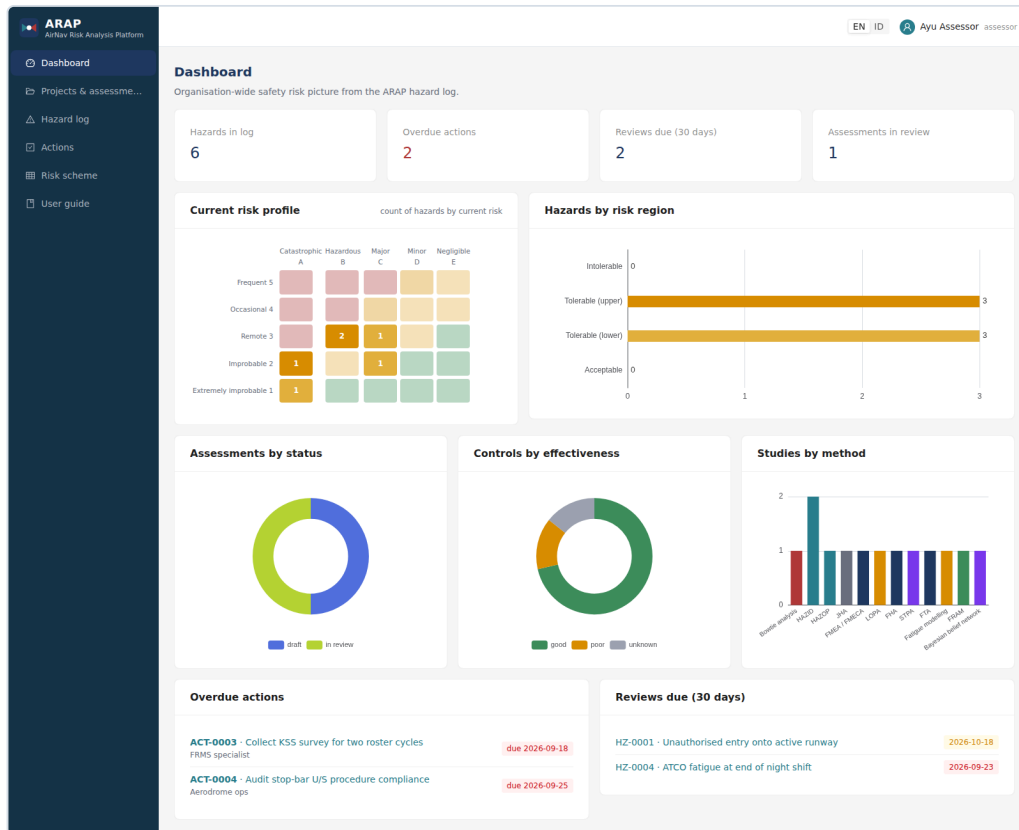
The administrator account is `admin` ; its password is set at installation. **Disable the demo data and change all passwords before real use.**

## 1.4 Screen layout and language

- The **left menu** gives access to the Dashboard, Projects & assessments, Hazard log, Actions, Risk scheme and User guide (plus Users and Audit trail for administrators and reviewers).
- The **EN / ID** switch at the top right changes the interface language between English and Bahasa Indonesia. Your choice is remembered on that computer. Text you type into worksheets is stored as entered.
- Your name and role are shown top right; click them to sign out.

## 2. The dashboard

The dashboard is the organisation-wide risk picture, built from the hazard log.



Dashboard

- **Key figures** — hazards in the log, overdue actions, hazard reviews due within 30 days, and assessments waiting for review.
- **Current risk profile** — the 5 × 5 risk matrix with the number of hazards in each cell (residual risk where rated, otherwise initial risk).
- **Hazards by risk region, Assessments by status, Controls by effectiveness** and **Studies by method** charts.
- **Overdue actions** and **Reviews due** lists — click an item to open it.

## 3. Projects and assessments

### 3.1 Projects

A **project** represents one change: a new system, a procedure, an airspace change, an organisational change or a temporary change. Go to **Projects & assessments** → **New project** and enter a code (e.g. ADSB-2027 ), a title, the type of change, the units affected and the sponsor.

**ARAP**  
AirNav Risk Analysis Platform

EN ID Ayu Assessor assessor

### Projects & assessments

A project is a change (system, procedure, airspace, organisational). Each has one or more safety assessments.

[+ New project](#)

#### Projects

| Code      | Title                                                 | Type           | Units                           | Assessments |
|-----------|-------------------------------------------------------|----------------|---------------------------------|-------------|
| APP-RELOC | Relocation of Jakarta approach unit to new ATC centre | organisational | Jakarta APP                     | 1           |
| DEMO-01   | Manual worked examples                                | system         | Jakarta ACC; Soekarno-Hatta TWR | 1           |

#### All assessments

| Project   | Assessment                           | Version | Status    | Updated          |
|-----------|--------------------------------------|---------|-----------|------------------|
| APP-RELOC | Transition safety assessment         | 1       | in review | 2026-09-28 10:22 |
| DEMO-01   | Worked examples — all twelve methods | 1       | draft     | 2026-09-28 10:22 |

< 1 >

Projects

### 3.2 Assessments

Open the project and choose **New assessment**. Describe the **scope and system boundary**, the **operational environment** and your **assumptions** (Manual §2.3). The assessment records which version of the risk scheme it uses.

The assessment page brings everything together:

**Worked examples — all twelve methods** draft v1

Submit for review | Report (.docx)

**Scope, environment and assumptions** Edit

Scope: Demonstration of every ARAP method module using the Manual's examples.  
 Environment: Illustrative values only.  
 Assumptions: All numbers are illustrative (Manual, How to use).  
 Risk scheme: version 1

**Method studies (12)** Add study

Suggested for a system change (Manual §4.1): HAZID FMEA FMEA / FMECA FTA Bowtie analysis

| Method                  | Study                                                           | Status | Updated          |
|-------------------------|-----------------------------------------------------------------|--------|------------------|
| Bowtie analysis         | Runway incursion bowtie (Manual §6)                             | open   | 2026-09-28 10:22 |
| HAZID                   | HAZID — approach unit relocation (Manual §7.4)                  | open   | 2026-09-28 10:22 |
| HAZOP                   | HAZOP — estimate message coordination (Manual §8.3)             | open   | 2026-09-28 10:22 |
| JHA                     | JHA — obstruction light on radar tower (Manual §9.3)            | open   | 2026-09-28 10:22 |
| FMEA / FMECA            | FMEA — sector VHF air-ground (Manual §10.3)                     | open   | 2026-09-28 10:22 |
| LOPA                    | LOPA — level bust (Manual §11.3)                                | open   | 2026-09-28 10:22 |
| FHA                     | FHA — ADS-B en-route surveillance (Manual §12.3)                | open   | 2026-09-28 10:22 |
| STPA                    | STPA — descent clearances (Manual §13.2)                        | open   | 2026-09-28 10:22 |
| FTA                     | FTA — loss of surveillance picture (Manual §14.3)               | open   | 2026-09-28 10:22 |
| Fatigue modeling        | Fatigue — night shift with/without nap (Manual §15.4)           | open   | 2026-09-28 10:22 |
| FRAM                    | FRAM — descent clearances in a busy arrival sector (Manual §16) | open   | 2026-09-28 10:22 |
| Bayesian belief network | BBN — coordination error and loss of separation (Manual §17.3)  | open   | 2026-09-28 10:22 |

**Hazards (5)** Open hazard log →

| Ref     | Hazard                                                   | Initial | Residual | Controls |
|---------|----------------------------------------------------------|---------|----------|----------|
| HZ-0001 | Unauthorised entry onto active runway                    | 3A      | 2A       | 2        |
| HZ-0002 | Level bust leading to potential mid-air collision        | 3A      | 3A       | 2        |
| HZ-0003 | Total loss of surveillance picture at ACC sector         | 3B      | 3B       | 1        |
| HZ-0004 | ATCO fatigue at end of night shift                       | 4C      | 3C       | 1        |
| HZ-0005 | New site outside radio line of sight of low-level sector | 3B      | —        | 0        |

**Risk acceptance**

Worst current/residual region: Tolerable (upper)  
 Acceptance authority: Accountable Executive

**Actions (5)** All actions →

- ACT-0001 Install diverse, physically separate surveillance data path  
 CNS engineering - due 2026-12-27 - open
- ACT-0002 Implement SFL mismatch alert in ATM system  
 ATM systems - due 2027-02-25 - open
- ACT-0003 Collect KSS survey for two roster cycles  
 FRMS specialist - due 2026-09-18 - open
- ACT-0004 Audit stop-bar U/S procedure compliance  
 Aerodrome ops - due 2026-09-25 - in progress
- ACT-0005 Radio coverage study for new site  
 CNS engineering - due 2026-10-28 - open

**Review and approval history**

Not yet submitted.

Assessment page

- **Method studies** — the analyses carried out for this assessment. ARAP suggests methods for the type of change (Manual §4.1); the suggestion is only advice.
- **Hazards** — hazard-log entries that belong to this assessment, with initial and residual risk.
- **Risk acceptance** — the worst residual risk region and the authority that must accept it.
- **Actions** and the **review and approval history**.
- Workflow buttons at the top right (see section 15) and **Report (.docx)**.

### 3.3 Adding a method study

On the assessment page choose **Add study**, pick the method (suggested methods are marked ★) and give the study a title, e.g. *HAZOP of cut-over plan*. The study opens straight away.

### 3.4 Starting from an assessment template

The screenshot shows the ARAP (AirNav Risk Analysis Platform) interface. A modal window titled "New safety assessment" is open. It has a close button (X) in the top right. Below the title, it says "Start from (optional)" and a dropdown menu is set to "New AIM system acquisition (installation → transition)".

Below the dropdown, there is an information icon (i) and a text box: "14 pre-filled studies will be created (Manual Appendix C)". Below this, it lists the studies: "FHA of AIM functions, data-flow HAZOP (7 nodes incl. migration), FMEA, FTA of undetected erroneous critical data, CCA, STPA, HTA + HRA, security, JHA and SWIFT of installation and cut-over, transition bowtie, shadow-operation plan and GSN skeleton." Below the list, there are buttons for each study type: FHA, HAZOP, FMEA, FTA, CCA, STPA, HTA, HRA, SEC, JHA, SWIFT, BOWTIE, SIM, and GSN. Below the buttons, it says "Content is proposed: review it, rate likelihoods and confirm severities in the workshops."

Below the studies section, there are three sections with text input fields:

- \* Title: "Safety assessment — new AIM system acquisition"
- \* Scope and system boundary (Manual §2.3): "Acquisition, installation, data migration and transition into operation of a new Aeronautical Information Management (AIM) system replacing the legacy AIS system. Functions in scope: receipt and registration of raw data from originators; validation and verification (incl. CRC-"
- Operational environment: "Aeronautical information for all Indonesian FIRs, published to the AIRAC 28-day cycle; NOTAM office operating H24; data integrity classification per ICAO Annex 15 and PANS-AIM"
- Assumptions: "A-1 The vendor provides software assurance evidence (EUROCAE ED-153 or equivalent) for the assigned assurance level. A-2 The legacy AIS system remains available as a hot-

At the bottom of the modal, there is a "Create" button.

*New assessment from a template*

In **New assessment**, the field **Start from** offers ready-made assessment templates. When you choose one:

- ARAP fills in the title, scope, environment and assumptions. Edit them for your change.
- On **Create**, ARAP also creates all of the template's method studies, pre-filled with proposed content.
- Nothing is rated for you. Likelihoods, calculations and hazard-log entries are left for your workshops, and severities are proposals to confirm.

**New AIM system acquisition** (Manual Appendix C) creates 14 studies covering the whole life of the change, from specification through installation and data migration to transition:

| Study                         | Pre-filled content                                                                                                                                                     |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FHA                           | 8 AIM functions and 12 failure conditions (loss, delay, detected and undetected error) with proposed severities and requirements                                       |
| Data-flow HAZOP               | 7 nodes from originator to user, including legacy → new migration; 15 deviations filled in (use <b>Deviations</b> to add the rest)                                     |
| FMEA                          | 8 architecture items: servers, fail-over, database, CRC module, validation rules, AMHS gateway, time source, eAIP web                                                  |
| FTA                           | "Erroneous critical data published undetected", with an entry-error branch and a storage/transfer corruption branch; data entry and verification come from the HRA     |
| CCA, STPA, HTA, HRA, security | Independence claims, the officer-automation control loop, the NOTAM publication task, data-entry and verification error probabilities, and integrity/access threats    |
| JHA, SWIFT                    | Installation works beside the live legacy AIS, and what-ifs for installation and cut-over                                                                              |
| Bowtie                        | "Loss of, or erroneous, AIS during transition", with migration, testing, training, interface, timing and installation threats, plus rollback and manual-NOTAM recovery |
| Simulation                    | The shadow-operation plan and acceptance criteria (zero critical-data differences; processing time; rejections)                                                        |
| GSN                           | The argument G0-G7, with every solution already linked to the new studies                                                                                              |

Demo project **DEMO-04** is this template filled in with illustrative values. It has likelihoods, a calculated fault tree ( $2.86 \times 10^{-5}$  per critical item, which is above the objective, so a further independent check is needed), shadow-operation results with one criterion not met, 7 hazards and 6 actions.

**Major organisational change** (Manual Appendix D) creates 11 studies covering planning, execution and post-implementation:

ARAP

AirNav Risk Analysis Platform

Dashboard

Projects & assessments

Hazard log

Actions

Risk scheme

User guide

EN ID

Ayu Assessor assessor

4 participants

Saved

Organisational function map

Organisational function map — current to new owner

Safety assessment — major organisational change · Manual chapter App. D · template v1.0

+ Add row

Delete selected

Send all rows to hazard log

Double-click a cell to edit. Tick rows to select.

| ID    |                          | Safety-related function                                                 | Gaps                             | Category   | Current owner                                | New owner                                         | Comp...   | Capacity   | Authority |
|-------|--------------------------|-------------------------------------------------------------------------|----------------------------------|------------|----------------------------------------------|---------------------------------------------------|-----------|------------|-----------|
| FN-01 | <input type="checkbox"/> | Accept residual risk in the Tolerable (lower) region for branch changes | authority pending                | SMS        | Director of Operations                       | Regional Operations Director                      | confirmed | adequate   | pending   |
| FN-02 | <input type="checkbox"/> | Own and review branch hazard-log entries                                | capacity stretched               | SMS        | Branch Safety Officer                        | Regional S&Q Unit                                 | confirmed | stretched  | in place  |
| FN-03 | <input type="checkbox"/> | Triage occurrence reports (ERC) within 72 h                             | capacity overloaded              | SMS        | Branch Safety Officer                        | Regional S&Q Unit                                 | confirmed | overloaded | in place  |
| FN-04 | <input type="checkbox"/> | Investigate branch occurrences                                          | competence<br>capacity stretched | SMS        | Branch Safety Officer (trained investigator) | Regional S&Q Unit                                 | gap       | stretched  | in place  |
| FN-05 | <input type="checkbox"/> | Safety Manager reports directly to the Accountable Executive            | authority none no handover       | SMS        | Safety Manager -- Accountable Executive      | Regional S&Q Unit -- Regional Operations Director | confirmed | adequate   | none      |
| FN-06 | <input type="checkbox"/> | Facilitate safety assessments of branch changes                         | capacity stretched               | SMS        | Branch Safety Officer                        | Regional S&Q Unit                                 | confirmed | stretched  | in place  |
| FN-07 | <input type="checkbox"/> | Supervise the APP watch                                                 | capacity stretched               | Operations | APP Watch Supervisor                         | Operations Supervisor (TWR + APP)                 | confirmed | stretched  | in place  |
| FN-08 | <input type="checkbox"/> | Supervise the TWR watch                                                 | capacity stretched               | Operations | TWR Watch                                    | Operations                                        | confirmed | stretched  | in place  |

18 functions mapped · 13 with gaps · 2 without a new owner · 2 still to assess

Every row needs a named new owner with confirmed competence, adequate capacity, authority in place and a planned handover before go-live. Send rows with gaps to the hazard log.

Organisational function map

| Study                       | Pre-filled content                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Organisational function map | 18 safety-related functions: SMS roles, supervision, training and competence, technical functions and bowtie barrier owners, each with current owner → new owner. Competence, capacity, authority and handover start as <i>unknown</i> . The <b>Gaps</b> column flags orphans (no new owner), competence gaps, stretched or overloaded capacity, pending authority, missing handover and anything still to assess. Rows with gaps can be sent to the hazard log. |
| FHA                         | Failure conditions of the organisational functions: triage delayed, investigations degraded, safety function not independent, merged supervisor overloaded, LoAs without an owner, and so on                                                                                                                                                                                                                                                                     |
| STPA × 2                    | The <b>current</b> and the <b>new</b> control structure, so you can compare the feedback loops. The new one includes unsafe control actions (UCAs) and loss scenarios.                                                                                                                                                                                                                                                                                           |
| HAZID                       | Uses its own <b>organisational guidewords</b> : roles, authority, competence, workload, interfaces, communication, corporate memory, contractors, culture, transition. Coverage is counted against this list.                                                                                                                                                                                                                                                    |
| HTA                         | The merged Operations Supervisor post, with a plan that sets priorities between watch, OJT and events                                                                                                                                                                                                                                                                                                                                                            |
| Delphi                      | The question and panel for the likelihood of triage failure after transfer                                                                                                                                                                                                                                                                                                                                                                                       |
| SWIFT, bowtie               | What-ifs for the transition plan; the top event "safety-related function not performed, or performed without competence, capacity or authority", with phased go/no-go and reversion                                                                                                                                                                                                                                                                              |
| SPI register                | 10 indicators with target, alert level, owner and escalation. Rows at red are listed at the top.                                                                                                                                                                                                                                                                                                                                                                 |
| GSN                         | G0–G7, linked to all the studies above                                                                                                                                                                                                                                                                                                                                                                                                                           |

**ARAP**  
AirNav Risk Analysis Platform

EN ID Ayu Assessor assessor

Dashboard
Projects & assessments...
Hazard log
Actions
Risk scheme
User guide

SPI register
**SPI register — post-implementation monitoring**

Safety assessment — major organisational change · Manual chapter 32 · template v1.0
4 participants
Saved

+ Add row
Delete selected
Double-click a cell to edit. Tick rows to select.

| ID   | Indicator                                                                                               | Type    | Barrier / hazard  | Base...      | Target ...   | Alert level     | Frequ...  | Owner                 | Current | Status |
|------|---------------------------------------------------------------------------------------------------------|---------|-------------------|--------------|--------------|-----------------|-----------|-----------------------|---------|--------|
| S-01 | <input type="checkbox"/> Vacant or acting posts among safety-critical functions                         | leading | Function map      | 0            | 0            | ≥ 2             | monthly   | HR                    | 1       | amber  |
| S-02 | <input type="checkbox"/> Function-map rows with gaps (owner, competence, capacity, authority, handover) | leading | Function map      | 18 to assess | 0 at go-live | ≥ 1 at go-live  | weekly    | Transition board      | 13      | red    |
| S-03 | <input type="checkbox"/> Overdue hazard-log actions and reviews (%)                                     | leading | SMS hazard log    | 6%           | ≤ 5%         | > 15%           | monthly   | Regional S&Q          | 9%      | amber  |
| S-04 | <input type="checkbox"/> Occurrence reports triaged within 72 h (%)                                     | leading | FHA triage        | 97%          | ≥ 95%        | < 85%           | monthly   | Regional S&Q          | 88%     | amber  |
| S-05 | <input type="checkbox"/> Voluntary safety reports per month vs baseline                                 | leading | Reporting culture | 24 / month   | ≥ baseline   | drop > 30%      | monthly   | Safety Manager        | 19      | amber  |
| S-06 | <input type="checkbox"/> Trainees waiting for OJT allocation                                            | leading | OJT coordination  | 2            | ≤ 3          | rising 2 months | monthly   | Terminal Unit Manager | 3       | green  |
| S-07 | <input type="checkbox"/> Overtime hours per ATCO per month                                              | leading | Fatigue / roster  | 8 h          | ≤ 10 h       | > 12 h          | monthly   | Terminal Unit Manager | 11 h    | amber  |
| S-08 | <input type="checkbox"/> Median time to close an investigation (days)                                   | lagging | Investigation     | 28           | ≤ 30         | > 60            | quarterly | Regional S&Q          | 41      | amber  |
| S-09 | <input type="checkbox"/> Safety culture survey index (0–100)                                            | leading | Culture           | 68           | ≥ 68         | drop ≥ 5 points | 6-monthly | Safety Manager        | —       |        |
| S-10 | <input type="checkbox"/> Occurrences with noncommunicational                                            | leading | Investigations    | 2 / quarter  | ≤ baseline   | rising 2        | quarterly | Regional S&Q          | 2       | amber  |

At alert level: S-02 — apply the agreed escalation

SPI register

Demo project **DEMO-05** is this template filled in for an illustrative branch reorganisation. It has 13 of 18 functions with gaps (2 orphans), the Safety & Quality function losing its direct line to the Accountable Executive, a Delphi estimate converging on 0.12, SPI values, 7 hazards and 6 actions.

The same modules can be added to any assessment on their own: **Organisational function map** and **SPI register** appear in **Add study**.

## 4. Working in a method study

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Every study page has the same header:

- The **method** tag and the **study title** (editable).
- A link back to the assessment, the Manual chapter for the method and the template version.
- The number of **participants** recorded.
- **Save**. Changes are not saved automatically: the button reads *Save* while there are unsaved changes and *Saved* afterwards. If you try to leave the page with unsaved changes ARAP asks you to confirm.

When the assessment is endorsed, accepted or closed, or if your role cannot edit, the study opens **read-only** and a blue banner says so.

### 4.1 Sending results to the hazard log

Every method has a **Send to hazard log** button. It creates a hazard-log entry for each selected worksheet row (or for the bowtie as a whole, or for each STPA hazard), carrying over the causes, consequences, controls and any severity and likelihood. Sending the same row again **updates** the existing hazard rather than creating a duplicate. A message lists the hazard references created, e.g. HZ-0007, HZ-0008 . Rate the risk and complete the controls in the [hazard log](#).

## 5. Worksheet methods

HAZID, HAZOP, JHA, FMEA/FMECA and FHA use a spreadsheet-style worksheet.

### Editing the worksheet

- **Double-click** a cell (or press Enter) to edit it; press Enter or click elsewhere to finish.
- Columns with fixed values — severity, likelihood, guideword, control type, failure type — show a drop-down list.
- **Add row** appends a row with the next ID ( HZ-06 , HP-04 , ...).
- Tick the checkboxes on the left to select rows, then **Delete selected** or **Send ... to hazard log**.
- Click a column header to sort; use the filter icon to filter.
- The **Risk** column is computed from severity and likelihood using the current risk matrix.

The screenshot displays the ARAP AirNav Risk Analysis Platform interface. On the left is a dark sidebar with navigation links: Dashboard, Projects & assessments, Hazard log, Actions, Risk scheme, and User guide. The main area shows a HAZID worksheet titled 'HAZID — approach unit relocation (Manual §7.4)'. At the top right of the main area, it says 'EN ID' and 'Ayu Assessor assessor'. Below the title, there's a note 'Worked examples — all twelve methods · Manual chapter 7 · template v1.0' and '2 participants' with a 'Saved' button. The worksheet table has columns: ID, Guideword, Hazard, Cause(s), Consequence, Existing controls, S, L, Risk, and Actions. There are five rows: HZ-01 (Transition, Loss of surveillance display at cut-over, Data feed not re-routed; configuration error, Loss of separation during cut-over, Cut-over at low traffic; old site on hot standby, C, 3, 3C, HAZOP of cut rollback criteria), HZ-02 (People, ATCOs unfamiliar with new CWP HMI, Short training window, Delayed or erroneous inputs, Simulator training, D, 4, 4D, Minimum hours before live ops), HZ-03 (Interfaces, Wrong telephone coordination lines to TWR, VCCS mapping error, Late coordination of missed approach, Test plan, C, 3, 3C, End-to-end line each TWR), HZ-04 (Equipment / systems, Single power feed to new building, Design, Total loss of service, UPS 30 min, B, 2, 2B, Second feed of power), and HZ-05 (Environment, New site outside radio line of sight of low-level sector, Topography, Loss of VHF contact below FL050, None yet, B, 3, 3B, Coverage study site). Below the table is a 'Guideword coverage' panel with a summary: People: 1, Procedures: 0, Equipment / systems: 1, Airspace and aerodrome: 0, Traffic: 0, Environment: 1, Interfaces: 1, Transition: 1, Organisation: 0. There is a dropdown menu to 'Record "no hazard identified" for a guideword'.

HAZID worksheet after sending two rows to the hazard log

### 5.1 HAZID (Manual §7)

Work through the guidewords (People, Procedures, Equipment/systems, Airspace and aerodrome, Traffic, Environment, Interfaces, Transition, Organisation). The **Guideword coverage** panel below the worksheet shows how many rows each guideword has. When a guideword produces nothing, record that explicitly with **Record “no hazard identified”**. Use the *Needs method* column to flag rows that need a detailed method.

### 5.2 HAZOP (Manual §8)

1. In **Nodes and design intent**, add a node per procedure step or message exchange (+ **Node**), choose its type (*procedural* or *data-flow*) and write the design intent.
2. Press **Deviations** on a node. Choose the parameters and guidewords; ARAP adds one worksheet row per combination (e.g. *Message — No / Not*).

- For each row either tick **N/A** (not meaningful) or record causes, consequences, safeguards, S, L and a recommendation.
- The **Progress** column counts rows analysed out of rows generated.

**Nodes and design intent**

| ID | Node                                          | Type      | Design intent                                                                                        | Progress                                |
|----|-----------------------------------------------|-----------|------------------------------------------------------------------------------------------------------|-----------------------------------------|
| N3 | Node 3: EST/CPL message upstream → downstream | data-flow | Downstream sector receives complete, correct and timely data for every flight crossing the boundary. | 3/3 analysed <a href="#">Deviations</a> |

+ Add row   Delete selected   Send all rows to hazard log   Double-click a cell to edit. Tick rows to select.

| ID    | Node | Parameter | Guideword  | Deviation                             | Possible causes                                        | Consequences                  | Safeguards                                    | S | L |
|-------|------|-----------|------------|---------------------------------------|--------------------------------------------------------|-------------------------------|-----------------------------------------------|---|---|
| HP-01 | N3   | Message   | No / Not   | Estimate not received                 | Link failure; flight plan not activated; address error | Unknown traffic enters sector | Link-status alarm; verbal coordination backup | C | 3 |
| HP-02 | N3   | Content   | Other than | Coordinated level differs from actual | Level changed after EST; no revision                   | Level conflict downstream     | Mode C/S check; STCA                          | B | 3 |
| HP-03 | N3   | Timing    | Late       | Estimate inside agreed time           | Short-notice route change                              | Workload                      | LoA fallback; verbal                          | D | 4 |

HAZOP worksheet with node panel

### 5.3 JHA (Manual §9)

Fill in the job header (job, location, permits), then one row per job step with its hazards, controls, control type and the person responsible. ARAP warns when a step relies only on **Administrative** controls or **PPE**, the lowest levels of the hierarchy of controls.

### 5.4 FMEA / FMECA (Manual §10)

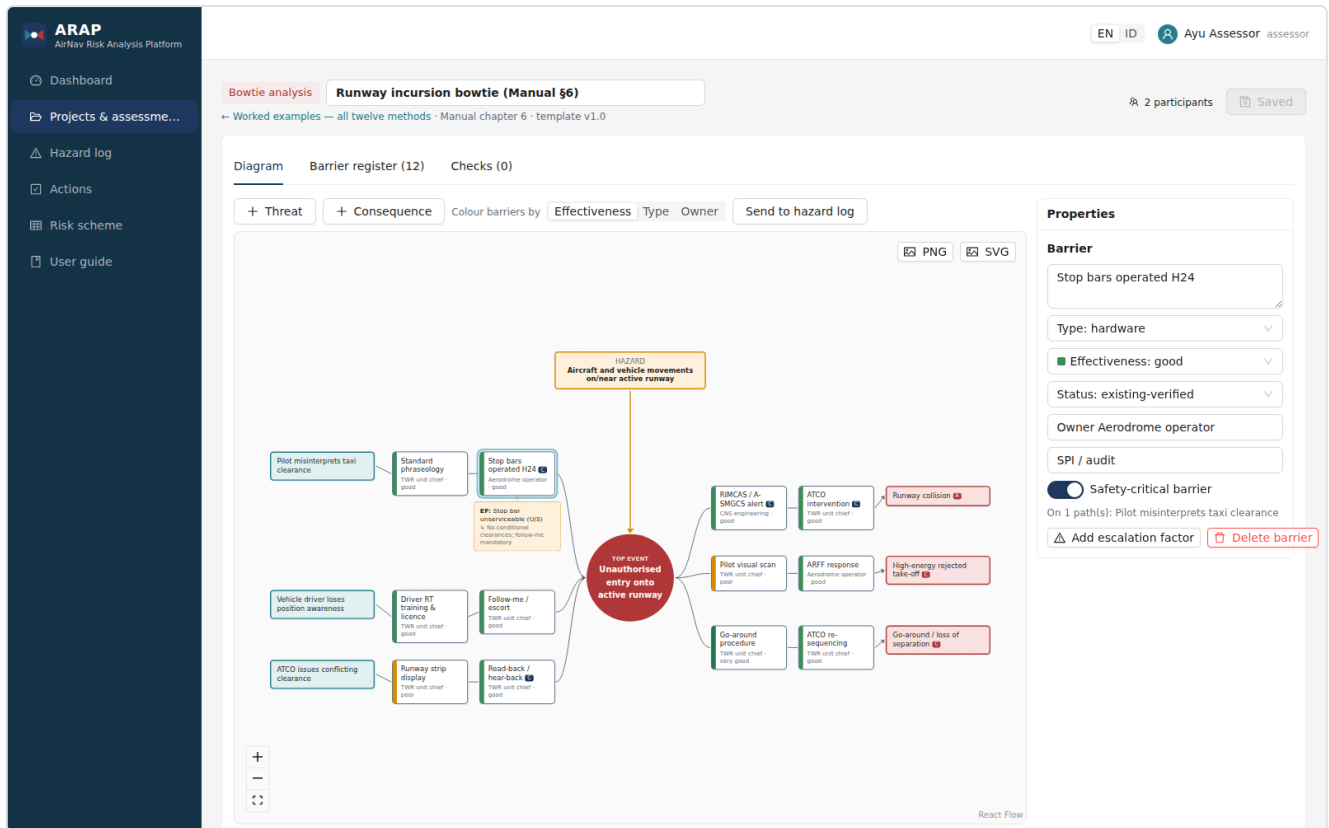
One row per failure mode: item, failure mode, cause, end effect on the ATS, detection, compensation, and **S, O, D** ratings (1–10). ARAP computes the **RPN** ( $S \times O \times D$ ), colours high values and sorts by RPN. It also lists high-severity modes ( $S \geq 8$ ) so they are not missed because of a low RPN. For FMECA, add the failure rate  $\lambda_p$ , mode ratio  $\alpha$ , conditional probability  $\beta$  and operating time  $t$ .

### 5.5 FHA (Manual §12)

One row per failure condition: the function, the failure type (total loss, partial loss, erroneous detected/undetected, unintended, delayed), the condition, the operational effect and the **severity**. The **Safety objective** column is computed from the risk scheme: the maximum frequency per operating hour that keeps the failure condition at or better than *Tolerable (lower)* (Manual Table 12.1). Record the derived safety requirements in the last column.

## 6. Bowtie analysis

The bowtie editor draws the diagram automatically: threats on the left, the top event in the centre, consequences on the right, barriers in order on each line and escalation factors below the barrier they weaken.



Bowtie editor with a barrier selected

### Building the bowtie

1. Click the **HAZARD** box and the **TOP EVENT** circle to edit their text. Phrase the top event as the moment control is lost, *before* any damage.
2. **+ Threat** and **+ Consequence** add lines. Click one to edit its text (and a consequence's severity).
3. With a threat or consequence selected, the **Properties** panel lists its barriers in order: - **New barrier on this path** adds a barrier; - **Add an existing (shared) barrier** puts a barrier that already exists on this line too — a shared barrier shows **xn**; - the arrows reorder barriers and the bin removes a barrier from this line only.
4. Click a **barrier** to set its type, effectiveness, status (existing-verified, existing-unverified, planned), owner, SPI/audit reference and whether it is **safety-critical** (C badge).
5. With a barrier selected, **Add escalation factor** attaches a condition that degrades it. Select the escalation factor to write its escalation-factor barriers, one per line.

### Viewing

- **Colour barriers by** effectiveness (default), type or owner.
- Zoom with the mouse wheel or the **+** / **–** controls; drag the background to pan; the frame icon fits the diagram.
- **PNG** and **SVG** download the full diagram as an image.
- **Barrier register** lists every barrier with its attributes (Manual §6.3).

- **Checks** lists issues: a top event phrased like a consequence, lines with fewer than two barriers, barriers without an owner.

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Dashboard
Projects & assessments...
Hazard log
Actions
Risk scheme
User guide

EN ID

Ayu Assessor assessor

Bowtie analysis

Runway incursion bowtie (Manual §6)

Worked examples — all twelve methods · Manual chapter 6 · template v1.0

2 participants

Saved

Diagram

Barrier register (12)

Checks (0)

| Barrier                      | Side       | Type           | Owner              | Effectiveness | Critical | Status            | Paths                                   | SPI / audit |
|------------------------------|------------|----------------|--------------------|---------------|----------|-------------------|-----------------------------------------|-------------|
| Standard phraseology         | prevention | human          | TWR unit chief     | good          |          | existing-verified | Pilot misinterprets taxi clearance      |             |
| Stop bars operated H24       | prevention | hardware       | Aerodrome operator | good          | Yes      | existing-verified | Pilot misinterprets taxi clearance      |             |
| Driver RT training & licence | prevention | human          | TWR unit chief     | good          |          | existing-verified | Vehicle driver loses position awareness |             |
| Follow-me / escort           | prevention | human          | TWR unit chief     | good          |          | existing-verified | Vehicle driver loses position awareness |             |
| Runway strip display         | prevention | human-hardware | TWR unit chief     | poor          |          | existing-verified | ATCO issues conflicting clearance       |             |
| Read-back / hear-back        | prevention | human          | TWR unit chief     | good          | Yes      | existing-verified | ATCO issues conflicting clearance       |             |
| RIMCAS / A-SMGCS alert       | recovery   | hardware       | CNS engineering    | good          | Yes      | existing-verified | Runway collision                        |             |
| ATCO intervention            | recovery   | human          | TWR unit chief     | good          | Yes      | existing-verified | Runway collision                        |             |
| Pilot visual scan            | recovery   | human          | TWR unit chief     | poor          |          | existing-verified | High-energy rejected take-off           |             |
| ARFF response                | recovery   | human-hardware | Aerodrome operator | good          |          | existing-verified | High-energy rejected take-off           |             |
| Go-around procedure          | recovery   | human          | TWR unit chief     | very good     |          | existing-verified | Go-around / loss of separation          |             |
| ATCO re-sequencing           | recovery   | human          | TWR unit chief     | good          |          | existing-verified | Go-around / loss of separation          |             |

Barrier register

## 7. LOPA

Risk scheme

User guide

Scenario Level bust leading to a potential mid-air collision

Initiating event Level bust in sector X

f(IE) 0.5

Unit per year

Data source Illustrative

Tolerable target 0.00001

Enabling conditions / conditional modifiers

+ Add

| Condition                   | Probability |
|-----------------------------|-------------|
| Conflicting traffic present | 0.05        |

Safeguards — credited only if all four IPL criteria are met

+ Add

| Safeguard                        | PFD  | indep.                              | effec.                              | depen.                              | audit.                              | Shared dependencies | Justi |
|----------------------------------|------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|---------------------|-------|
| ATCO monitoring and intervention | 0.10 | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | surveillance        |       |
| STCA alert and ATCO resolution   | 0.10 | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | surveillance        |       |
| ACAS II RA followed              | 0.10 | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | e.g. surveillance   |       |
| SFL mismatch alert (proposed)    | 0.10 | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | e.g. surveillance   |       |
| Recurrent RT training            | 0.10 | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | e.g. surveillance   |       |

Calculate

Mitigated frequency

2.50e-5 per year

Target

1.00e-5 per year

Verdict

Gap x2.5

Further reduction needed

x2.5 (0.40 orders of magnitude)

Credited IPLs

ATCO monitoring and intervention; STCA alert and ATCO resolution; ACAS II RA followed

Not credited: SFL mismatch alert (proposed)

Missing: dependable

Not credited: Recurrent RT training

Missing: independent, effective

IPLs ATCO monitoring and intervention, STCA alert and ATCO resolution share dependency 'surveillance' — independence must be justified

| Layer                            | Frequency (per year) |
|----------------------------------|----------------------|
| Initiating event                 | 5.0e-1               |
| Conflicting traffic present      | 2.5e-2               |
| ATCO monitoring and intervention | 2.5e-3               |
| STCA alert and ATCO resolution   | 2.5e-4               |
| ACAS II RA followed              | 2.5e-5               |

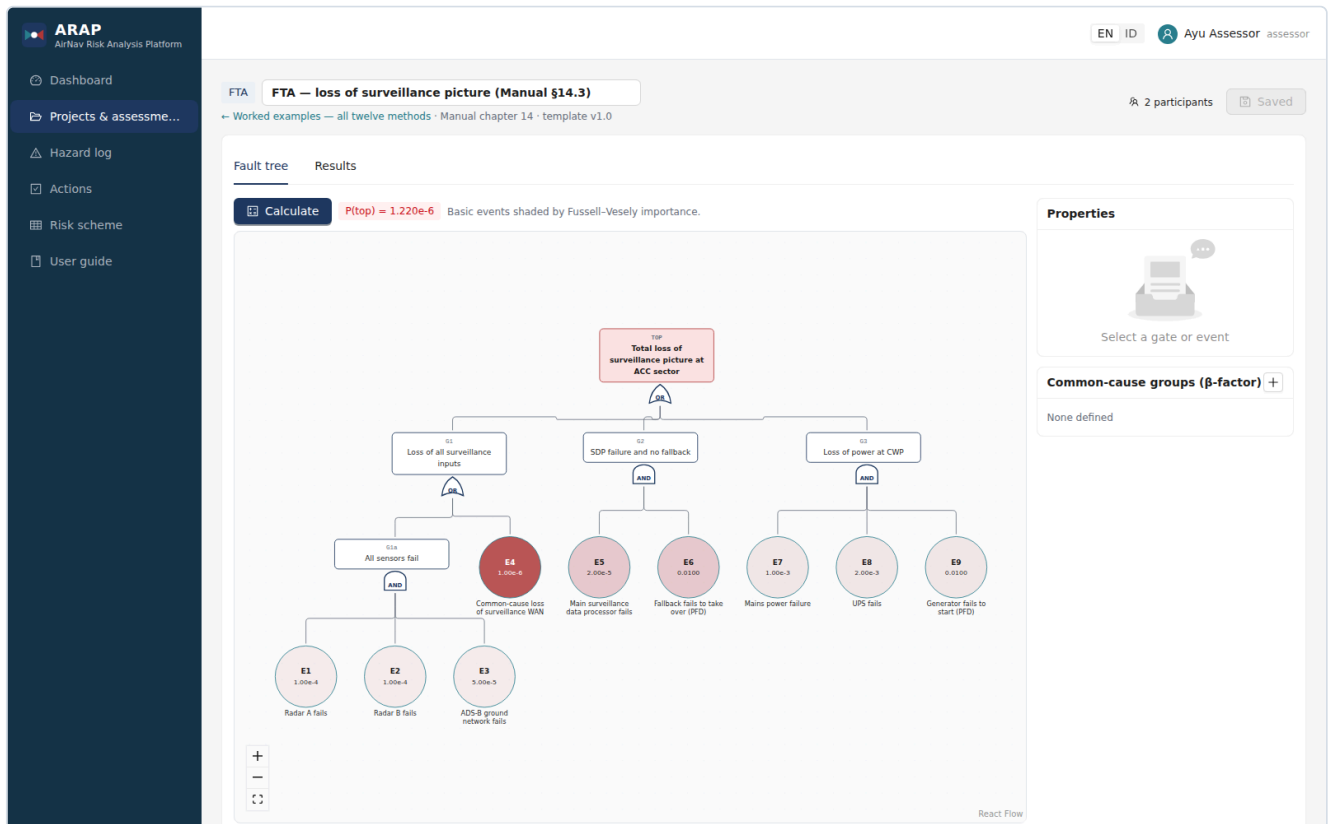
Engine lopa-1.0.0. LOPA is an order-of-magnitude screening tool (Manual §11.4).

LOPA calculator

1. Describe the **scenario**, the **initiating event**, its frequency, unit and data source, and the **tolerable target** frequency.
2. Add **enabling conditions / conditional modifiers** with their probabilities.
3. Add **safeguards** with a probability of failure on demand (PFD). Tick each of the four IPL criteria — *independent, effective, dependable, auditable* — only if it is justified, and write the justification. A safeguard is credited **only if all four are ticked**.
4. Record shared dependencies (e.g. **surveillance**) — ARAP warns if two credited layers share one.
5. Press **Calculate**. The result shows the mitigated frequency, the verdict (*Target met* or *Gap x n*), the extra risk reduction needed, the safeguards not credited and why, and a log-scale chart of the frequency after each layer.

In the demo study the proposed *SFL mismatch alert* is not yet ticked *dependable*, so the result shows a gap of x2.5. Tick it and recalculate to see the target met ( $2.5 \times 10^{-6}$  per year).

## 8. Fault tree analysis



Fault tree

### Building the tree

- Click a gate and use **OR gate**, **AND gate**, **Basic event** or **Undeveloped** in the Properties panel to add inputs. **Reuse an existing node as input** links a node already in the tree (e.g. a shared support system).
- For a **vote** gate set  $k$  ( $k$ -out-of- $n$ ).
- For a **basic event** give either a probability  $p$ , or a failure rate  $\lambda$  with an exposure time  $t$  ( $P = 1 - e^{-\lambda t}$ ), a repair time MTTR (repairable unavailability), or a test interval  $T$  (periodically tested standby,  $\lambda T/2$ ).
- A **house event** is a switch set to TRUE or FALSE.
- Common-cause groups** apply the  $\beta$ -factor model to a set of basic events.

### Calculating

Press **Calculate**. The top-event probability appears above the tree, and basic events are shaded by their Fussell-Vesely importance. The **Results** tab gives:

- the exact top-event probability (binary decision diagram), the rare-event approximation and the min-cut upper bound;
- single points of failure;
- minimal cut sets with probability and share;
- Birnbaum and Fussell-Vesely importance for every basic event.

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Dashboard

Projects & assessments

Hazard log

Actions

Risk scheme

User guide

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FTA — loss of surveillance picture (Manual §14.3)

Worked examples — all twelve methods · Manual chapter 14 · template v1.0

2 participants

Save

Fault treeResults

|                                    |                          |
|------------------------------------|--------------------------|
| Top-event probability (exact, BDD) | 1.2200e-6                |
| Rare-event approximation           | 1.2200e-6                |
| Min-cut upper bound                | 1.2200e-6                |
| Single points of failure           | E4                       |
| Engine                             | fta-1.1.0 · 31 BDD nodes |

Minimal cut sets

| Cut set      | Order | Probability | Share |
|--------------|-------|-------------|-------|
| {E4}         | 1     | 1.00e-6     | 82.0% |
| {E5, E6}     | 2     | 2.00e-7     | 16.4% |
| {E7, E8, E9} | 3     | 2.00e-8     | 1.6%  |
| {E1, E2, E3} | 3     | 5.00e-13    | 0.0%  |

Importance measures

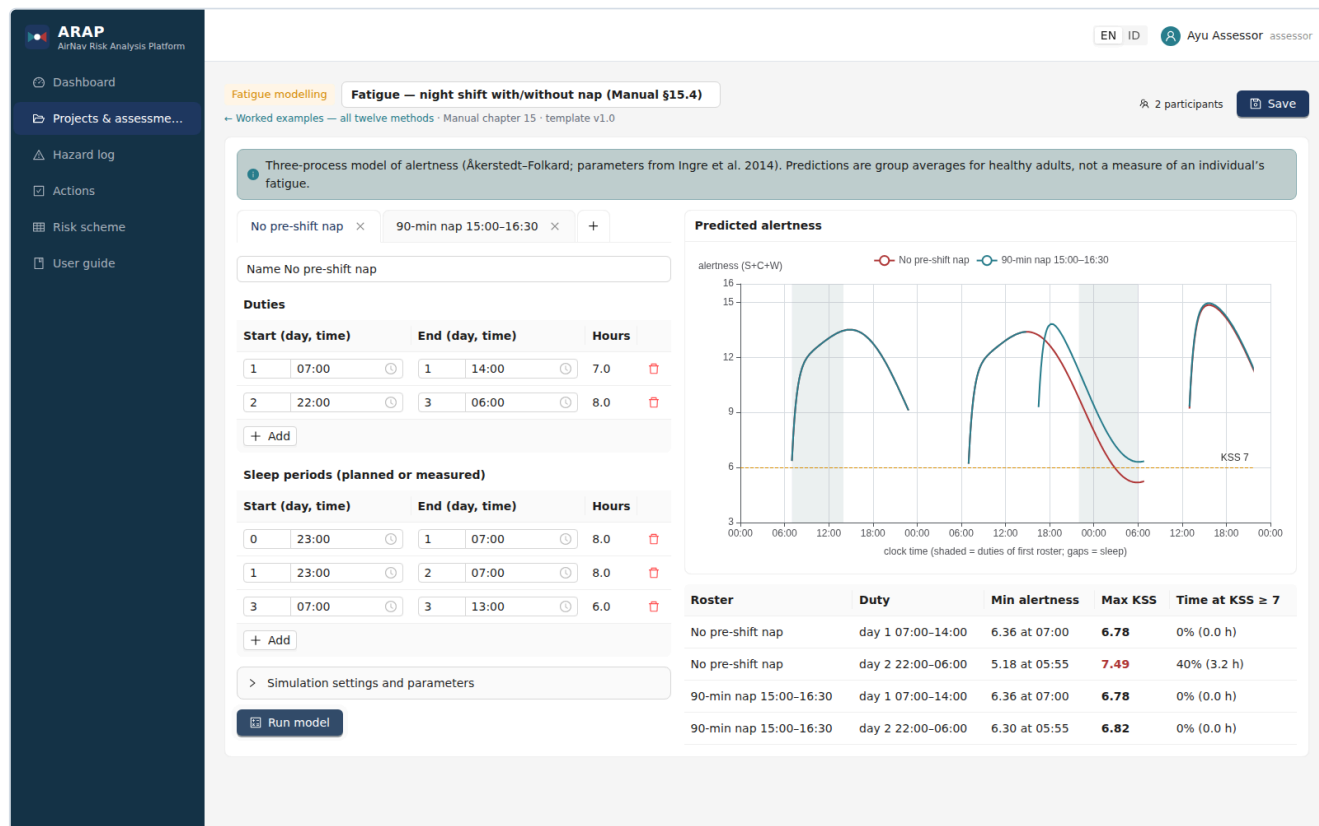
| Event | Description                            | p       | Birnbaum | Fussell-Vesely |
|-------|----------------------------------------|---------|----------|----------------|
| E4    | Common-cause loss of surveillance WAN  | 1.00e-6 | 1.00     | 0.820          |
| E5    | Main surveillance data processor fails | 2.00e-5 | 1.00e-2  | 0.164          |
| E6    | Fallback fails to take over (PFD)      | 0.0100  | 2.00e-5  | 0.164          |
| E7    | Mains power failure                    | 1.00e-3 | 2.00e-5  | 0.0164         |
| E8    | UPS fails                              | 2.00e-3 | 1.00e-5  | 0.0164         |
| E9    | Generator fails to start (PFD)         | 0.0100  | 2.00e-6  | 0.0164         |
| E1    | Radar A fails                          | 1.00e-4 | 5.00e-9  | 4.098e-7       |
| E2    | Radar B fails                          | 1.00e-4 | 5.00e-9  | 4.098e-7       |
| E3    | ADS-B ground network fails             | 5.00e-5 | 1.00e-8  | 4.098e-7       |

FTA results

ARAP User Guide · v0.5

19 / 48

## 9. Fatigue modelling

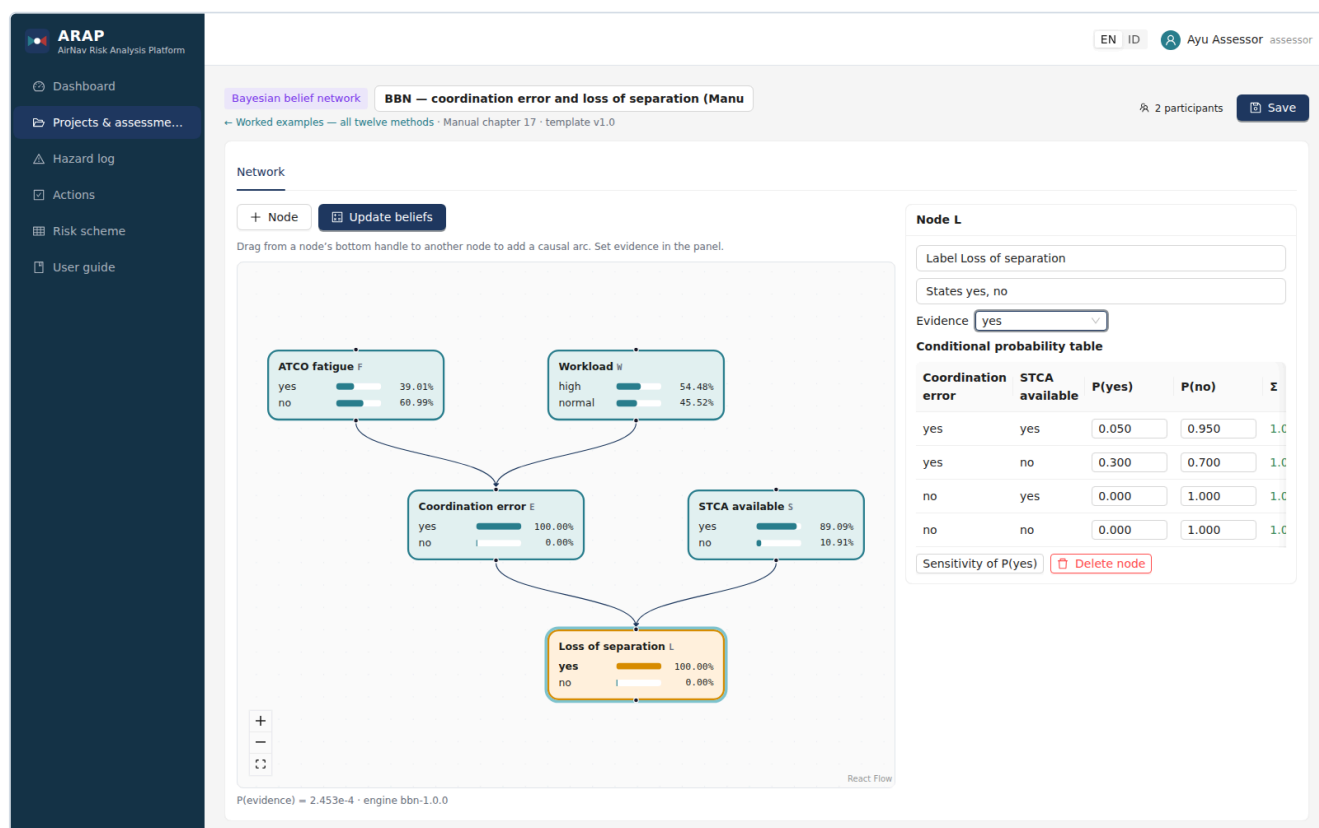


Fatigue model

1. Each tab is a **roster variant** (e.g. *No pre-shift nap* and *90-min nap*). Use **+** to copy the first roster into a new variant.
2. Enter **duties** and **sleep periods** as day number and clock time. Day 1 starts at 00:00; a sleep starting at 23:00 the evening before day 1 is entered as day 0, 23:00.
3. **Simulation settings and parameters** sets the number of days, time step, KSS threshold and, if needed, the model parameters (leave blank for the defaults of the Manual, §15.2).
4. Press **Run model**. The chart shows predicted alertness for each variant (gaps are sleep; shaded bands are duties of the first variant; the dashed line is the KSS threshold). The table gives, per duty, the minimum alertness and when it occurs, the maximum KSS, and the time spent at or above the threshold.

Predictions are group averages for healthy adults. They are not a measure of an individual's fatigue and must be checked against operational data (Manual §15.5).

## 10. Bayesian belief networks



Bayesian network with evidence

### Building the network

- **+ Node** adds a node; drag nodes to arrange them.
- To add a causal arc, drag from the **bottom handle** of the cause to the effect node. Double-click an arc to remove it. ARAP rejects cycles when calculating.
- Select a node to edit its label and **states** (comma-separated) and its **conditional probability table** — one row per combination of parent states. Each row must sum to 1; the Σ column turns red when it does not (for two-state nodes the second value fills in automatically).

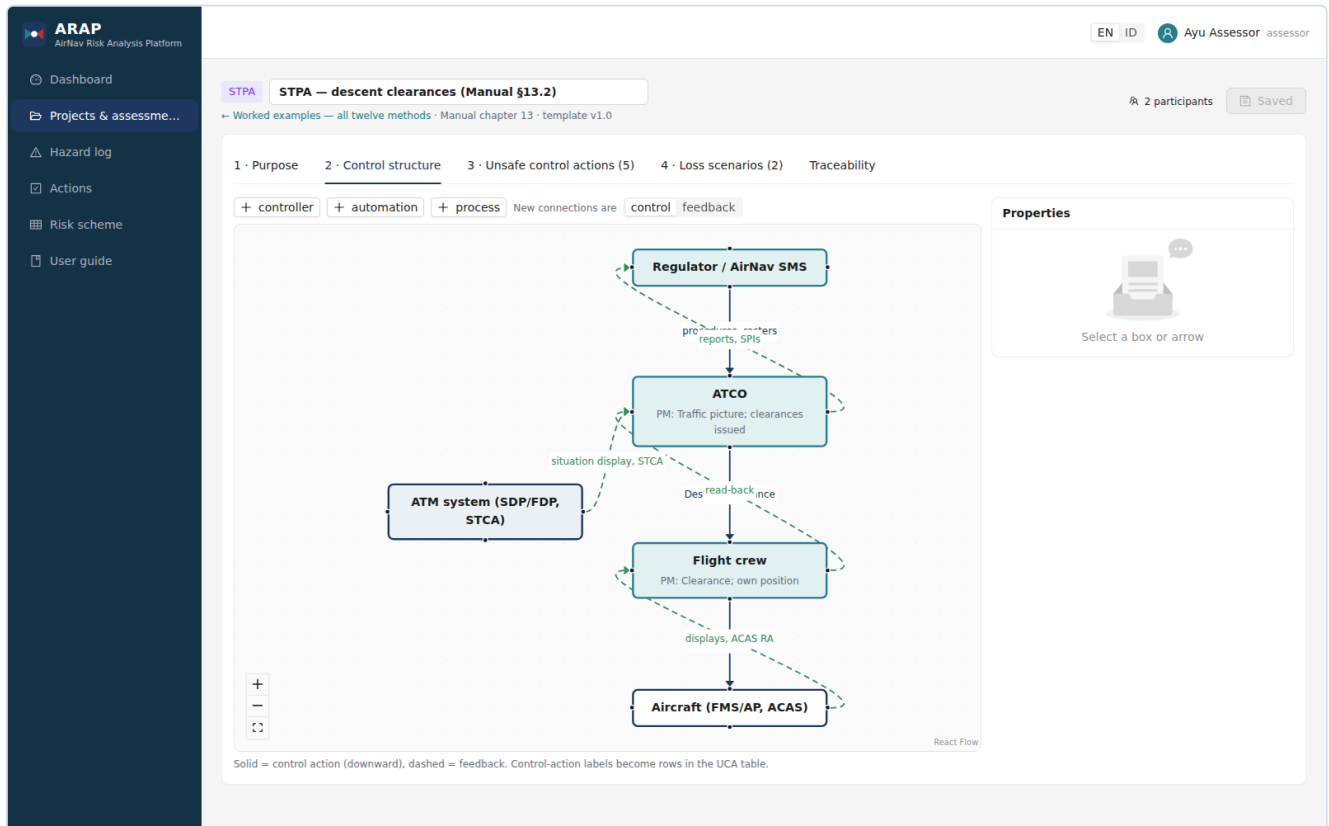
### Using the network

- **Update beliefs** calculates every node's probabilities; the bars inside the nodes show them.
- Select a node and choose a state under **Evidence** to condition on it (the node turns amber). All other nodes update — this is how you ask diagnostic questions such as *given a loss of separation, how likely was fatigue?*
- **Sensitivity** shows how much each other node can move the selected node's probability (tornado chart).

## 11. STPA

The STPA study follows the four steps of the STPA Handbook, one tab each.

1. **Purpose** — losses, system-level hazards (linked to losses) and system-level constraints (linked to hazards).
2. **Control structure** — add controllers, automation and controlled processes; drag from a box's bottom/right handle to another box to add an arrow. Choose whether new arrows are **control** actions (solid) or **feedback** (dashed). Click a box to write its *process model*; click an arrow to rename it.



STPA control structure

3. **Unsafe control actions** — for every control action, four columns (*not providing, providing, too early/too late/wrong order, stopped too soon/applied too long*). Add UCAs with **+ UCA**; each needs a context ("when ...") and at least one hazard, otherwise ARAP warns.

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Projects & assessme...

Hazard log

Actions

Risk scheme

User guide

STPA — descent clearances (Manual §13.2)

Worked examples — all twelve methods · Manual chapter 13 · template v1.0

2 participants

Saved

1 · Purpose

2 · Control structure

3 · Unsafe control actions (5)

4 · Loss scenarios (2)

Traceability

Control action: procedures, rosters

Not providing causes hazard

Providing causes hazard

Too early / too late / wrong order

Stopped too soon / applied too long

+ UCA

+ UCA

+ UCA

+ UCA

Control action: Descent clearance

Not providing causes hazard

Providing causes hazard

Too early / too late / wrong order

Stopped too soon / applied too long

UCA-1

ATCO does not issue descent

when the aircraft must leave its level to avoid a conflict ahead

H-1 ×

remove

+ UCA

UCA-2

ATCO clears descent through an occupied level

when unseparated traffic is at that level

H-1 ×

remove

UCA-3

ATCO clears descent below MVA

when vectoring over high terrain

H-2 ×

remove

+ UCA

UCA-6

Descent issued before preceding aircraft vacated level

when relying on the label trend

H-1 ×

remove

+ UCA

+ UCA

Control action: Execute descent

Not providing causes hazard

Providing causes hazard

Too early / too late / wrong order

Stopped too soon / applied too long

+ UCA

+ UCA

+ UCA

UCA-10

Descent issued through cleared

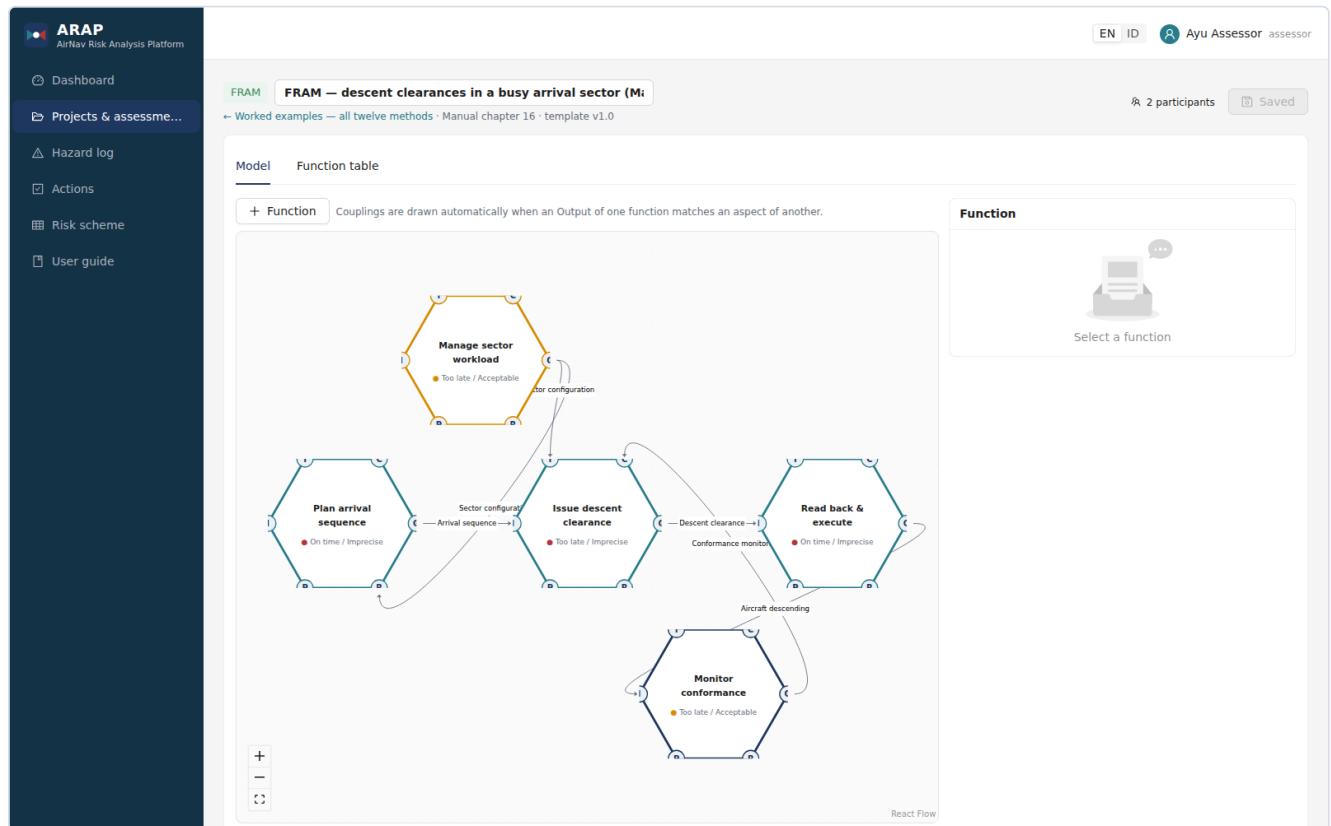
UCA table

- Loss scenarios** — for each UCA, the scenario type, description and the resulting requirement or constraint. **Send hazards to hazard log** creates one hazard-log entry per STPA hazard.
- Traceability** — the chain loss → hazard → constraint/UCA → scenario → requirement, with broken chains flagged.

ARAP User Guide · v0.5

23 / 48

## 12. FRAM



FRAM model

- **+ Function** adds a function (hexagon). Drag to arrange.
- Select a function to set its name, type (human, technological, organisational) and the six **aspects** — Input, Output, Precondition, Resource, Control, Time. Type an entry and press Enter; existing outputs are offered as suggestions.
- **Couplings are drawn automatically** wherever an Output of one function has the same text as an aspect of another. Unmatched aspects are listed so you can fix typos or add missing functions.
- Record the **output variability** (timing: too early / on time / too late / not at all; precision: precise / acceptable / imprecise). The dot colour on each hexagon shows it.
- The **Function table** tab lists everything in one table.

## 13. Extended methods (Manual chapters 18–31)

Version 0.2 adds thirteen methods, and version 0.3 adds the wildlife strike module. They all work like the others: open the study, edit, press **Calculate** where there is one, and **Save**. The demo project **DEMO-02** has one worked example of each, matching the Manual.

### 13.1 Collision risk — CRM (Manual §18)

The screenshot displays the ARAP AirNav Risk Analysis Platform interface. The left sidebar contains navigation links: Dashboard, Projects & assessments, Hazard log, Actions, Risk scheme, and User guide. The main content area is titled 'Collision risk modelling' and 'CRM — RVSM vertical risk and parallel-route spacing'. It includes a note about the Reich collision risk model and two tabs: 'Vertical (RVSM)' and 'Lateral (route spacing)'. The 'Lateral (route spacing)' tab is active, showing a 'Parameters' section with input fields for aircraft dimensions, lateral deviation model, route spacing, and various occupancy and speed parameters. A 'Calculate' button is at the bottom of the parameters section. To the right, the 'Result' section shows a table with calculated values and a log-linear plot of collision risk versus route spacing, with a vertical line indicating the minimum spacing of 11.7 NM.

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Collision risk modelling CRM — RVSM vertical risk and parallel-route spacing

Worked examples — thirteen extended methods · Manual chapter 18 · template v1.0

2 participants Save

Reich collision risk model in occupancy form (ICAO Doc 9689 / Doc 9574). Overlap probabilities and occupancies should come from regional monitoring agency data (e.g. height-keeping and lateral deviation monitoring).

Vertical (RVSM) Lateral (route spacing)

**Parameters**

$\lambda_x$  length (ft) 184  $\lambda_y$  wingspan (ft) 175

$\lambda_z$  height (ft) 50

Lateral deviation model Double expone... scale (NM) 0.9

Route spacing  $S_y$  (NM) 15

$P_z(0)$  — vertical overlap, same level 0.52  $E_y(\text{same})$  — same-direction lateral occupancy 0.1

$E_y(\text{opp})$  — opposite-direction lateral occupancy 0  $S_x$  — longitudinal window (NM) 80

$|\Delta v|$  (kt) 20  $|v|$  (kt) 470

$|z|$  (kt) 1.5  $|y(S_y)|$  lateral closing speed at loss of spacing (kt) 4

Target level of safety 0.000000005

Calculate

**Result**

|                                     |                     |
|-------------------------------------|---------------------|
| $P_y(S_y)$ at 15 NM                 | 1.63e-8             |
| Lateral risk (fatal accidents / fh) | 1.578e-10 Meets TLS |
| Minimum spacing meeting TLS         | 11.68 NM            |

collision risk / fh

route spacing (NM)

min 11.7 NM

CRM lateral spacing

Two tabs. **Vertical (RVSM)** takes the Reich-model parameters: overlap probabilities, occupancies, speeds and aircraft dimensions (entered in feet). **Calculate** shows the same- and opposite-direction risk, the total and whether it meets the target level of safety (TLS). **Lateral (route spacing)** also asks for the lateral-deviation model (double exponential or Gaussian) and its scale. It plots risk against route spacing on a log scale and gives the **minimum spacing** that meets the TLS. The demo gives  $1.87 \times 10^{-9}$  (vertical) and 11.7 NM (lateral).

## 13.2 Event tree analysis — ETA (Manual §19)

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Projects & assessments

Hazard log

Actions

Risk scheme

User guide

Event tree analysis

ETA — conflict after a level bust (Manual §19.3)

Worked examples

thirteen extended methods

Manual chapter 19

template v1.0

2 participants

Save

Initiating event

Conflict following a level bust

Frequency 0.025 per year

A successful barrier ends the sequence (barrier event tree)

Barriers / functional events (in order)

| ID    | Barrier                        | P(success) |
|-------|--------------------------------|------------|
| B1    | ATCO detects and resolves      | 0.90       |
| B2    | STCA alert and ATCO resolution | 0.90       |
| B3    | ACAS II RA followed            | 0.90       |
| + Add |                                |            |

Conditional probabilities (depend... path prefix e.g. "F" or "F-F")

| After path | Barrier | P(success) |
|------------|---------|------------|
| F          | B2      | 0.80       |
| + Add      |         |            |

Calculate

Initiating event

Conflict following a level bust

B1

ATCO detects and resolve

B2

STCA alert and ATCO reso

B3

ACAS II RA followed

D

Resolved by ATCO — no loss of separation

S

0.0225 per year

E

Loss of separation, resolved after STC

F-S

2.00e-3 per year

F

Serious loss of separation, resolved by ACAS

F-F-S

4.50e-4 per year

A

Near mid-air collision / collision

F-F-F

5.00e-5 per year

Up (green) = barrier succeeds / down (red) = barrier fails

Outcomes

| Sequence | Outcome                                      | Severity | Probability | Frequency (per year) |
|----------|----------------------------------------------|----------|-------------|----------------------|
| S        | Resolved by ATCO — no loss of separation     | D        | 0.900       | 0.0225               |
| F-S      | Loss of separation, resolved after STCA      | C        | 0.0800      | 2.000e-3             |
| F-F-S    | Serious loss of separation, resolved by ACAS | B        | 0.0180      | 4.500e-4             |
| F-F-F    | Near mid-air collision / collision           | A        | 2.000e-3    | 5.000e-5             |

Σ probability = 1.000000 · by severity: D 0.0225 · C 2.00e-3 · B 4.50e-4 · A 5.00e-5 · edit outcomes then recalculate

Event tree

1. Enter the initiating event and its frequency.
2. List the barriers **in the order they act**, with their probability of success.
3. Under **Conditional probabilities**, enter any dependence as a path prefix. For example, **F / B2 / 0.8** means "STCA succeeds 80% of the time when the controller has already failed".
4. Press **Calculate**. ARAP draws the tree and lists each sequence with its probability and frequency.
5. Name each outcome and give it a severity, then recalculate. The totals by severity appear under the table.

## 13.3 Human reliability — HRA (Manual §20)

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Dashboard

Projects & assessments...

Hazard log

Actions

Risk scheme

User guide

EN ID

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Human reliability (HEART / CARA)

HRA — controller and technician tasks (Manual §20.4)

2 participants

Save

HEP =  $GTT \times \prod[(EPC - 1) \times APOA + 1]$ . Use CARA for controller tasks and HEART for other tasks (e.g. engineering). Justify every APOA; use the HEP in FTA, LOPA or ETA.

Tasks

| ID | Task                                                                     | GTT     | HEP      |
|----|--------------------------------------------------------------------------|---------|----------|
| T1 | ATCO acts correctly on an STCA alert                                     | CARA B3 | 2.016e-3 |
| T2 | ATCO detects a level bust by monitoring the display (no alert)           | CARA B1 | 0.0575   |
| T3 | Technician restores radar to operational configuration after maintenance | HEART F | 0.0108   |

Calculate all (server)

Task T1

ATCO acts correctly on an STCA alert

CARA (ATC)

HEART

B3 — Respond to unique and trusted audible and visual i...

Error-producing conditions

| EPC | Condition          | Max | APOA        | Factor |
|-----|--------------------|-----|-------------|--------|
| 17  | Traffic complexity | ×10 | <div></div> | 2.80   |
| 6   | Cognitive overload | ×6  | <div></div> | 1.50   |
| 13  | Low vigilance      | ×3  | <div></div> | 1.20   |

Add an error-producing condition

Nominal HEP 0.0004 → assessed HEP 2.016e-3

Justification of the task type and APOA values

HRA

Add a task and choose **CARA** (controller tasks) or **HEART** (other tasks) and the generic task type. Add error-producing conditions from the list. For each one, set the **APOA** slider (0–1) and write a justification. The HEP updates as you edit. **Calculate all** runs the server engine for the record.

## 13.4 Occurrence risk classification — ERC and RAT (Manual §21)

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Occurrence risk classification Occurrence risk classification — ERC and RAT (Manual §21)

← Worked examples — thirteen extended methods · Manual chapter 21 · template v1.0

2 participants Saved

**Occurrences** + Occurrence

| ID      | Occurrence                | ERC | RAT | Class |
|---------|---------------------------|-----|-----|-------|
| OCC-101 | Level bust FL240→FL233... | 102 | 7   | B     |
| OCC-102 | Vehicle entered runway... | 50  | —   | —     |
| OCC-103 | Frequency congestion ...  | 1   | —   | —     |

Send red-band occurrences to hazard log

**OCC-101**

Level bust FL240→FL233 with crossing traffic; STCA; ATCO resolved; 2.8 NM / 700 ft

Select date ESARR 2 severity class (analyst) B

**ARMS ERC** **RAT scoring**

**Minimum separation achieved**  
Separation ≥ minimum (0) > 75% of minimum (1) > 50% and ≤ 75% (3) > 25% and ≤ 50% (7) ≤ 25% of minimum (10)

**Rate of closure**  
None (0) Low (≤ 85 kt, ≤ 1000 ft/min) (1) Medium (≤ 205 kt, ≤ 2000 ft/min) (2) High (≤ 700 kt, ≤ 4000 ft/min) (4) Very high (> 700 kt, > 4000 ft/min) (5)

**Conflict detection (ATM ground)**  
Detected (0) Detected late (1) Not detected (2) Not applicable (0)

**Planning (ATM ground)**  
Plan correct (0) Plan inadequate (1) No plan (2) Not applicable (0)

**Execution**  
Execution correct (0) Execution inadequate (1) No execution (2) Not applicable (0)

**Ground safety net (STCA / MSAW / APW)**  
Triggered (0) Not triggered (2) Not applicable (0)

**Recovery**  
Recovery correct (0) Recovery inadequate (1) No recovery / worsened (2) Not applicable (0)

**Airborne safety net / pilot initiative**  
TCAS/GPWS or see-and-avoid effective (0) No RA / warning (2) Not applicable (0)

**Risk of collision 5 + controllability 2 = severity score 7**  
Assign the ESARR 2 severity class with the EUROCONTROL RAT look-up table (not embedded in ARAP) and record it above.

Notes / rationale

### ERC

- **Add an occurrence.** Add a short description, date and occurrence ID.
- **ARMS ERC.** Click the matrix cell that answers Q1 (most credible accident outcome) and Q2 (effectiveness of the remaining barriers). The band and recommended response are shown.
- **RAT scoring.** Pick one answer per item. ARAP adds the points for risk of collision and controllability. Record the ESARR 2 severity class that you read from the EUROCONTROL RAT table.
- **Send red-band occurrences to hazard log.** Creates hazard entries for the red-band occurrences.

## 13.5 Safety argument — GSN (Manual §22)

ARAP

AirNav Risk Analysis Platform

Dashboard

Projects & assessments...

Hazard log

Actions

Risk scheme

User guide

EN ID

Ayu Assessor assessor

Safety case (GSN)

Safety argument — ADS-B introduction (Manual §22.3)

2 participants

Saved

Structure valid

6 goals

4/4 solutions linked to evidence

1 undeveloped

Solid arrow = SupportedBy · open arrow, dashed = InContextOf · ◊ = undeveloped. Click an element to edit.

Sn4 — solution

Real-time simulation results (workload, R/T occupancy)

Parent | G6

Evidence

ARAP study

#22 SIM — Simulation — new arrival procedu...

Status: **SIM #22 · open** · open study

Evidence is not yet complete / accepted — the argument is provisional.

Argument checks

No structural issues found.

GSN

- **Build the argument.** Click an element to edit it. Use **+ goal** / **+ strategy** / **+ solution** / **+ context** / **+ assumption** / **+ justification** to add elements below it. ARAP lays out the diagram automatically.
- **Link evidence.** For a **solution**, link the evidence: an ARAP study, a hazard log entry or an external document. The solution shows the evidence status in green (complete or accepted) or red (still provisional).
- **Argument checks.** This panel lists structural problems: more than one top goal, unsupported goals, strategies supported by something other than goals, solutions without evidence, and cycles. Mark a goal **undeveloped** when its support is still to come.

## 13.6 Common cause analysis — CCA (Manual §23)

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Dashboard

Projects & assessments

Hazard log

Actions

Risk scheme

User guide

Every AND / vote gate in this FTA already has an independence claim

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Common cause analysis Common cause analysis — ACC surveillance chain (Man)

Worked examples — thirteen extended methods · Manual chapter 23 · template v1.0

2 participants Save

Every AND gate or redundancy claim in an FTA, RBD or FHA rests on independence. CCA checks it three ways: where items are installed (ZSA), what single events can hit several at once (PRA) and what they share by design (CMA).

3 independence claim(s) not supported 0 claim(s) not yet assessed 3 particular risk(s) defeat redundancy Send failures of independence to hazard log

Zonal safety analysis (3) Particular risks (4) Common mode analysis (3)

+ Add row Delete selected Import claims from FTA

| ID    | Independence claim                                                                  | Items claimed indepe... | Common-mode...                          | Analysis                                                      | Indepen... | Action                                     |
|-------|-------------------------------------------------------------------------------------|-------------------------|-----------------------------------------|---------------------------------------------------------------|------------|--------------------------------------------|
| CM-01 | <input type="checkbox"/> Main SDP (E5) and fallback (E6) fail independently         | E5, E6                  | Software                                | Fallback runs the same software baseline as the main system   | No         | Diverse fallback or $\beta$ -factor in FTA |
| CM-02 | <input type="checkbox"/> Mains (E7), UPS (E8) and generator (E9) fail independently | E7, E8, E9              | Location                                | UPS and generator switchgear share the power room (fire zone) | Partly     | Include fire CCF in FTA                    |
| CM-03 | <input type="checkbox"/> Radar A, radar B and ADS-B inputs are independent          | E1, E2, E3              | Shared utilities (power, HVAC, network) | All three use the single surveillance WAN (E4)                | No         | Diverse data path (FTA shows 82% of risk)  |

CCA

Three worksheets: **Zonal safety analysis**, **Particular risks** and **Common mode analysis**. On the common-mode tab, **Import claims from FTA** creates one independence claim for every AND or k-out-of-n gate in a chosen fault tree. For each claim, record the common-mode source and whether independence holds. The tags at the top count unsupported claims and particular risks that defeat redundancy. **Send failures of independence to hazard log** passes them on.

## 13.7 RBD and Markov (Manual §24)

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Projects & assessme...

Hazard log

Actions

Risk scheme

User guide

RBD and Markov analysis

RBD and Markov — sector VHF availability (Manual §24.)

2 participants

Save

Reliability block diagram

Markov model

series: Sector frequency available at CWP

parallel: Radio chains

Main Tx/Rx chain

MTBF 20000 h · MTTR 4 h

A = 0.999800

Standby Tx/Rx chain

MTBF 20000 h · MTTR 4 h

A = 0.999800

parallel: Links to remote site

Leased line

MTBF 4000 h · MTTR 8 h

A = 0.998004

VSAT backup

MTBF 3000 h · MTTR 12 h

A = 0.996016

VCCS

MTBF 50000 h · MTTR 2 h

A = 0.999960

parallel: Power

Mains

MTBF 2000 h · MTTR 2 h

A = 0.999901

UPS

MTBF 30000 h · MTTR 8 h

A = 0.999733

Calculate availability

Blocks are assumed independent — check with common cause analysis.

|                   |                       |
|-------------------|-----------------------|
| Availability      | 0.99995174            |
| Unavailability    | 4.826e-5              |
| Expected downtime | 0.423 h/yr (25.4 min) |

VCCS

Leased line

VSAT backup

Mains

UPS

Main Tx/Rx chain

Standby Tx/Rx chain

downtime saved if perfect (min/yr)

0 5 10 15 20 25

Selected element

Label Sector frequency available at CWP

series parallel koon block

Add inside: + block + series + parallel + koon

RBD

- **RBD tab.** Click a group or block to edit it. Groups can be **series**, **parallel** or **k-out-of-n**. Blocks have MTBF and MTTR. Add elements inside a group with the + buttons.
- **Calculate availability.** Gives availability, unavailability and downtime per year, plus a chart of how many minutes a year each block would save if it were perfect.
- **Markov model tab.** Define states (tick *Service up?*) and transitions with rates per hour, then press **Solve**. You get steady-state probabilities, availability, failure frequency, system MTBF, mean down time and mean time to first failure.

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Dashboard

Projects & assessme...

Hazard log

Actions

Risk scheme

User guide

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RBD and Markov analysis

RBD and Markov — sector VHF availability (Manual §24.)

2 participants

Save

Reliability block diagram

Markov model

Use a Markov model where blocks are not independent: shared repair crews, cold/warm standby, imperfect switchover (coverage). Rates are per hour ( $\lambda = 1/\text{MTBF}$ ,  $\mu = 1/\text{MTTR}$ ).

States

| ID    | Description                    | Service up?                         |
|-------|--------------------------------|-------------------------------------|
| S2    | Main in service, standby ready | <input checked="" type="checkbox"/> |
| S1    | On standby, main in repair     | <input checked="" type="checkbox"/> |
| S0    | No radio chain in service      | <input type="checkbox"/>            |
| + Add |                                |                                     |

Transitions

| From  | To | Rate (/h)   | Meaning                      |
|-------|----|-------------|------------------------------|
| S2    | S1 | 0.000049504 | main fails, switchover OK    |
| S2    | S0 | 0.000000504 | main fails, switchover fails |
| S1    | S0 | 0.00005     | standby fails                |
| S1    | S2 | 0.25        | main repaired                |
| S0    | S1 | 0.25        | one chain restored           |
| + Add |    |             |                              |

Initial state (for MTTF)

Solve

|                   |                         |
|-------------------|-------------------------|
| Availability      | 0.999997960             |
| Unavailability    | 2.0396e-6 (1.07 min/yr) |
| Failure frequency | 5.099e-7 /h             |
| System MTBF       | 1,961,176 h             |
| Mean down time    | 4.00 h                  |
| MTTF              | 1,961,565 h (223.9 yr)  |

| State | Description                    | $\pi$ (steady state) |
|-------|--------------------------------|----------------------|
| S2    | Main in service, standby ready | 1.00                 |
| S1    | On standby, main in repair     | 1.9996e-4            |
| S0    | No radio chain in service      | 2.0396e-6            |

Markov

## 13.8 SWIFT (Manual §25)

A worksheet like HAZID. Each row is a *what if...?* question under a prompt category, with consequence, safeguards, severity, likelihood and recommendation. **Prompt category coverage** shows which categories have been considered. Use it to record "considered — nothing credible" for the rest. Rows with a consequence can be sent to the hazard log.

ARAP User Guide · v0.5

32 / 48

## 13.9 Hierarchical task analysis — HTA (Manual §26)

HTA

- **Build the hierarchy.** Select a task and press **Sub-task** to add a step below it. Numbering is automatic (1, 1.1, 1.2 ...).
- **Add plans.** Write a **plan** for every task that has sub-tasks. Tasks without one are flagged in red.
- **Record error modes.** Add error modes to bottom-level tasks.
- **Link to HRA.** Choose a **linked HRA study** at the top, then link a task to one of its HRA tasks to show the HEP.
- **Change the view.** The **Outline** tab shows the same content as an indented table.

## 13.10 Simulation (Manual §27)

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Dashboard

Projects & assessments

Hazard log

Actions

Risk scheme

User guide

EN ID

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Simulation (RTS / FTS)

Simulation — new arrival procedure (Manual §27.4)

2 participants Save

Worked examples — thirteen extended methods · Manual chapter 27 · template v1.0

Exercise plan

Title RTS of new arrival procedure (sector APP-N)

Real-time simulation

Safety objectives / hypotheses

Show the new procedure does not increase workload or R/T occupancy and does not add conflicts.

Scenarios, traffic samples, runs and design

Peak traffic 38 arrivals/h; 8 runs per condition; 4 ATCO teams; counterbalanced order

Participants (ATCOs, pseudo-pilots)

8 current APP ATCOs, 4 pseudo-pilots

Limitations and validity (realism, learning effects)

Measures and run data

one value per run, comma-separated

| ID | Measure              | Unit | Better | Criterion                        | Value | Baseline |
|----|----------------------|------|--------|----------------------------------|-------|----------|
| M1 | Workload (ISA, 1-5)  | ISA  | lower  | threshold                        | 3.5   | 3.4, 3   |
| M2 | STCA alerts per hour | /h   | lower  | no worse than baseline (+margin) | 0     | 1.2, 0   |
| M3 | R/T occupancy        | %    | lower  | threshold                        | 60    | 58, 63   |

+ Add

Analyse

Results

All success criteria met.

| Measure                                | Baseline mean ± sd (n) | Solution mean ± sd (n) | Solution 95% CI | Δ     | Welch's t-test |
|----------------------------------------|------------------------|------------------------|-----------------|-------|----------------|
| M1 Workload (ISA, 1-5), executive ATCO | 3.69 ± 0.20 (8)        | 3.16 ± 0.17 (8)        | 3.02 – 3.30     | -0.52 | <0.001         |
| M2 STCA alerts per hour                | 1.10 ± 0.23 (8)        | 0.85 ± 0.16 (8)        | 0.72 – 0.98     | -0.25 | 0.021          |
| M3 R/T occupancy                       | 61.63 ± 2.67 (8)       | 51.75 ± 2.12 (8)       | 49.98 – 53.52   | -9.88 | <0.001         |

% of baseline mean

Baseline Solution Solution 95% CI

Simulation

- **Plan the exercise.** Record the exercise plan: objectives, scenarios, participants and limitations.
- **Define the measures.** For each measure, set whether lower or higher is better and a **criterion**. Use *threshold* (e.g.  $\leq 3.5$ ), or *no worse than baseline*, which takes an optional margin. Set the criteria **before** entering results.
- **Enter the run results.** Enter them as comma-separated numbers for the baseline and the solution.
- **Analyse.** You get the mean, standard deviation, 95% confidence interval, Welch's t-test and whether each criterion is met. You can send failed criteria to the hazard log.

## 13.11 Expert judgement (Manual §28)

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Dashboard

Projects & assessme...

Hazard log

Actions

Risk scheme

User guide

Structured expert judgement

Expert judgement — ATCO detection probability (Manu...

2 participants

Saved

Classical model (Cooke)

Delphi

1

Cooke's classical model: each expert gives 5/50/95% quantiles for seed questions (true value known to the facilitator) and target questions. Weights = calibration × information; experts below the significance level  $\alpha$  get zero weight.

Expert name

+ Expert

+ Question

$\alpha$  0.00

4 seed · 1 target

Calculate weights

| ID | Question                                                   | Seed ?                              | Realisation | Scale  | Expert A |      |     | Expert B |     |     | Expert C |     |     |
|----|------------------------------------------------------------|-------------------------------------|-------------|--------|----------|------|-----|----------|-----|-----|----------|-----|-----|
|    |                                                            |                                     |             |        | 5%       | 50%  | 95% | 5%       | 50% | 95% | 5%       | 50% | 95% |
| S1 | Level busts reported in sector X last year                 | <input checked="" type="checkbox"/> | 14          | linear | 8        | 12   | 18  | 5        | 10  | 30  | 20       | 25  | 35  |
| S2 | STCA alerts per 1000 flights (sector X)                    | <input checked="" type="checkbox"/> | 2.1         | linear | 1        | 2    | 3   | 0.5      | 1.5 | 5   | 3        | 4   | 6   |
| S3 | Median ATCO response time to STCA (s)                      | <input checked="" type="checkbox"/> | 9           | linear | 6        | 9    | 14  | 4        | 8   | 20  | 12       | 15  | 20  |
| S4 | Read-back errors per 1000 clearances                       | <input checked="" type="checkbox"/> | 3.5         | linear | 2        | 3.5  | 6   | 1        | 4   | 10  | 6        | 8   | 12  |
| Q1 | P(ATCO does not resolve a level-bust conflict before STCA) | <input type="checkbox"/>            | target      | log    | 0.03     | 0.08 | 0.2 | 0.01     | 0.1 | 0.4 | 0.1      | 0.2 | 0.4 |

Expert scores

| Expert   | Calibration | Information | Bins    | Weight |
|----------|-------------|-------------|---------|--------|
| Expert A | 0.839       | 0.771       | 0/2/2/0 | 0.814  |
| Expert B | 0.596       | 0.249       | 0/1/3/0 | 0.186  |
| Expert C | 2.54e-5     | 0.562       | 4/0/0/0 | 0.000  |

Decision maker (combined distribution)

| Item                                                  | Performance-weighted 5 / 50 / 95% | Equal-weight 5 / 50 / 95% |
|-------------------------------------------------------|-----------------------------------|---------------------------|
| S1 seed<br>Level busts reported in sector X last year | 5.86 / 11.9 / 26.1                | 5.66 / 16.3 / 33.6        |
| S2 seed<br>STCA alerts per 1000 flights (sector X)    | 0.658 / 1.97 / 4.35               | 0.626 / 2.68 / 5.73       |

### Expert judgement

- Classical model (Cooke).** 1. Add experts and questions. 2. Tick **Seed?** for calibration questions and enter the true value. 3. Enter each expert's 5%, 50% and 95% values. Quantiles that are incomplete or not increasing are flagged. 4. Press **Calculate weights** to get calibration, information and weight per expert, and the combined (decision-maker) distribution for every question.
- Delphi.** Add panellists and rounds and enter the estimates, then press **Summarise rounds** to see the median, quartiles and convergence per round.

## 13.12 Security risk (Manual §29)

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Dashboard

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Hazard log

Actions

Risk scheme

User guide

Security risk assessment

Security risk assessment — surveillance and FDP (Mani

2 participants

Saved

Worked examples — thirteen extended methods · Manual chapter 29 · template v1.0

+ Add row

Delete selected

Send all rows to hazard log

Double-click a cell to edit. Tick rows to select.

| I.    |                          | Asset                            | Threat scenario                                        | Threat source          | Vulnerability                               | C | I | A | L... | Security risk | Existing controls                       |
|-------|--------------------------|----------------------------------|--------------------------------------------------------|------------------------|---------------------------------------------|---|---|---|------|---------------|-----------------------------------------|
| SR-01 | <input type="checkbox"/> | ADS-B ground station / SDP input | Injection of false ADS-B targets (spoofing)            | External attacker      | ADS-B messages unauthenticated              | 1 | 5 | 3 | 3    | 15 Very high  | Radar cross-check available; multilater |
| SR-02 | <input type="checkbox"/> | Flight data processing (FDP)     | Ransomware encrypts FDP servers                        | External attacker      | Flat network between office and ops systems | 3 | 4 | 5 | 2    | 10 High       | Backups; antivirus                      |
| SR-03 | <input type="checkbox"/> | VCCS maintenance access          | Compromise of vendor remote-maintenance account        | Supplier / third party | Shared vendor credentials                   | 3 | 4 | 5 | 2    | 10 High       | VPN                                     |
| SR-04 | <input type="checkbox"/> | ATM adaptation data              | Unauthorised change to sector or safety-net parameters | Insider (accidental)   | No two-person rule on adaptation changes    | 1 | 5 | 2 | 2    | 10 High       | Change log                              |

Security risk =  $L \times \max(C, I, A)$ . High / very high: SR-01, SR-02, SR-03, SR-04. 4 threat scenario(s) have a safety effect — send those to the safety hazard log (ED-205 / Manual §29.3).

### Security risk

A worksheet of threat scenarios: asset, threat, source, vulnerability, C/I/A impact (1–5), likelihood (1–5), controls, treatment and owner. **Security risk** = likelihood × the highest of C, I and A, shown as Low / Medium / High / Very high. Give each scenario a **safety effect** (severity A–E) where it has one. Only those rows are sent to the safety hazard log, so exploit details stay in the security study.

## 13.13 Occurrence investigation — SOAM, HFACS, Tripod Beta (Manual §30)

ARAP

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Dashboard

Projects & assessments

Hazard log

Actions

Risk scheme

User guide

Occurrence investigation

Investigation OCC-102 — works vehicle on runway strip

2 participants

Saved

Worked examples — thirteen extended methods · Manual chapter 30 · template v1.0

Occurrence SOAM HFACS Tripod Beta

Organisational factors

Training (TR)

Contractor drivers not trained on runway closure procedures

Change management (CM)

Works plan changed the closure time without a risk assessment

Communication (CO)

Revised closure time not briefed to contractor

Contextual conditions

Workplace conditions

Night works; poor signage visibility near K3

Human performance limitations

Driver's mental model: runway closed for works

Physiological and emotional factors

Driver at end of 12-hour shift

Human involvement

Driver crossed the lit stop bar believing the runway was closed

TWR did not challenge the driver's position report

Absent / failed barriers

Awareness - failed

Driver RT training and licence

Restriction - absent

Follow-me / escort for works vehicles

Restriction - failed

Stop bar at holding point K3

Occurrence

OCC-102

At night a works vehicle entered the runway strip of 07L without clearance. RIMCAS alerted; the TWR ATCO instructed an aircraft on 4 NM final to go around. Closes

Barriers that worked

RIMCAS alert

ATCO go-around instruction

Barriers

link failed barriers to the bowtie they belong to

| Barrier                               | Type        | Status | Bowtie barrier                                      |
|---------------------------------------|-------------|--------|-----------------------------------------------------|
| Driver RT training and licence        | Awareness   | failed | PB3 Driver RT training & licence — Runway incursion |
| Follow-me / escort for works vehicles | Restriction | absent | PB4 Follow-me / escort — Runway incursion           |
| Stop                                  |             |        |                                                     |

Contextual conditions

| Category                            | Condition                                      |
|-------------------------------------|------------------------------------------------|
| Workplace conditions                | Night works; poor signage visibility near K3   |
| Human performance limitations       | Driver's mental model: runway closed for works |
| Physiological and emotional factors | Driver at end of 12-hour shift                 |
| + Add                               |                                                |

Organisational factors

| Factor                 | Finding |
|------------------------|---------|
| Contractor drivers not |         |

SOAM

- **Occurrence** — the facts, the counts of failed barriers, HFACS categories and actions, and **Send occurrence to hazard log**.
- **SOAM** — barriers (type and status), human involvement, contextual conditions, organisational factors and safety actions. The SOAM chart updates as you type. Link each failed or absent barrier to the **bowtie barrier** it corresponds to. The bowtie then shows "recorded as failed in N investigation(s)" when you select that barrier.
- **HFACS** — tick categories in the four tiers and record the evidence for each.
- **Tripod Beta** — event trios (agent, object, event) with failed or missing barriers traced to immediate cause, precondition, underlying cause and Basic Risk Factor. A BRF profile summarises them.

## 13.14 Wildlife strike risk — species matrix (Manual §31)

**Species risk matrix** rows: likelihood · columns: severity

| # | Species                                                      | Strikes/yr | Per 10k mvt | Damaging | L × S       | Risk     | Trend  |
|---|--------------------------------------------------------------|------------|-------------|----------|-------------|----------|--------|
| 1 | Cattle egret (kuntul kerbau) <i>Bubulcus ibis</i>            | 12.4       | 1.90        | 11.3%    | 5 × 4 = 20  | high     | ▼ 0.09 |
| 2 | Large flying fox (kalong besar) <i>Pteropus vampyrus</i>     | 0.6        | 0.09        | 33.3%    | 2 × 5* = 10 | moderate | ▲ 0.02 |
| 3 | Pacific golden plover (cereb kernyut) <i>Pluvialis fulva</i> | 2.2        | 0.34        | 9.1%     | 3 × 3 = 9   | moderate | ▼ 0.01 |
| 4 | Javan pond heron (blekok sawah) <i>Ardeola speciosa</i>      | 4.4        | 0.67        | 4.5%     | 4 × 2 = 8   | moderate | ▼ 0.06 |
| 5 | Barn swallow (layang-layang asia) <i>Hirundo rustica</i>     | 18.0       | 2.75        | 0.0%     | 5 × 1 = 5   | low      | ▼ 0.23 |
| 6 | Eurasian tree sparrow (burung gereja) <i>Passer montanus</i> | 8.0        | 1.22        | 0.0%     | 4 × 1 = 4   | low      | ▼ 0.06 |

1 high · 3 moderate · 2 low All species: 228 strikes, 6.97 per 10,000 movements

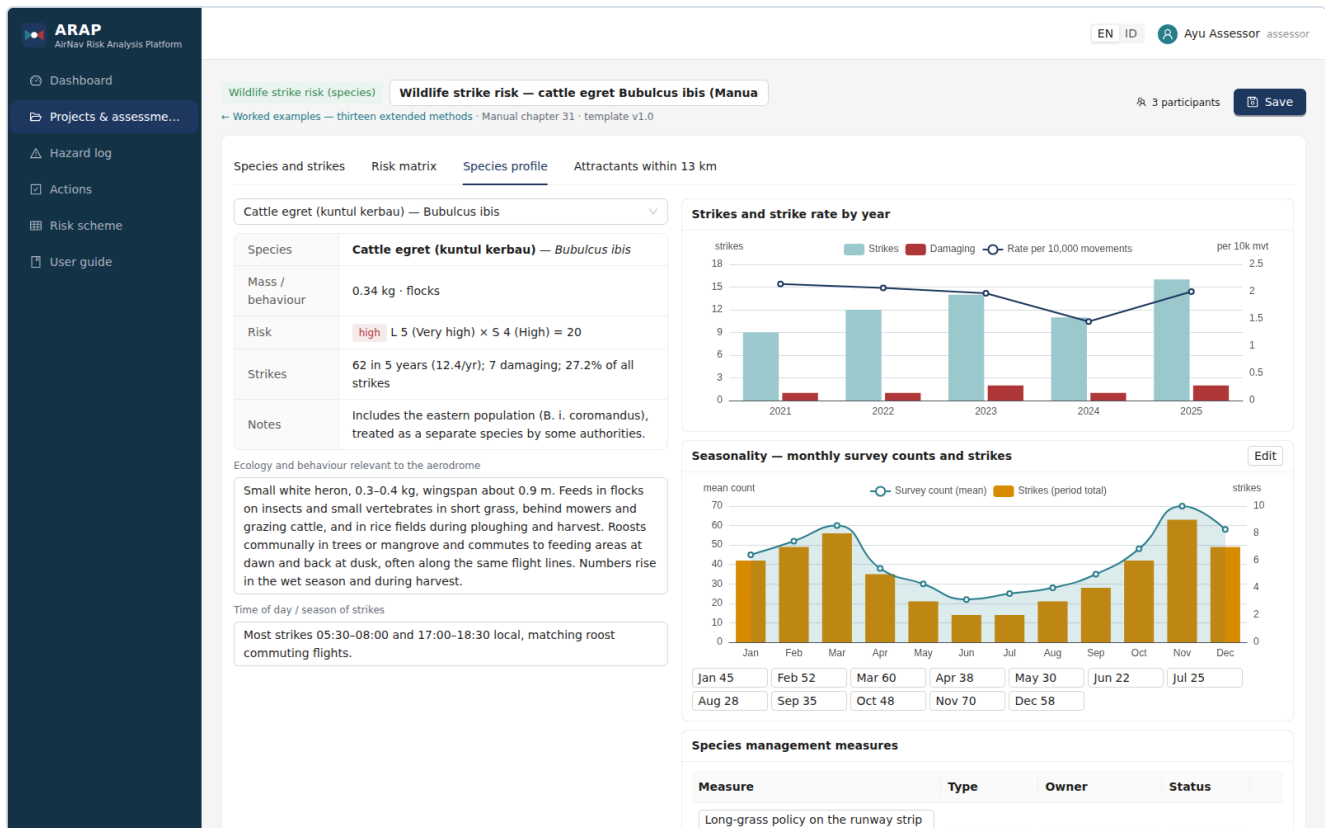
Send high and moderate species to hazard log

\* severity from the body-mass surrogate (too few strikes for a stable percentage). Engine wildlife-1.0.0.

Wildlife species risk matrix

- **Species and strikes.** Enter the review years and the aircraft movements for each year. Then add one row per species, or per group of similar species: common and scientific name, body mass, whether it flocks, and the strikes and damaging strikes for each year. Press **Calculate risk**.
- **Risk matrix.** Each species is placed on a 5 × 5 grid. Rows are likelihood (mean strikes per year) and columns are severity (share of strikes that caused damage or had an effect on flight). The ranking table adds the rate per 10,000 movements and its trend. A species with fewer than 5 strikes gets its severity from body mass, plus one class if it flocks; ARAP marks this with \*. Use **Send high and moderate species to hazard log** to create hazard entries, with the species' attractants and measures.
- **Species profile.** For the selected species: ecology notes, time of day of strikes, a year-by-year chart of strikes and strike rate, a monthly chart of survey counts and strikes (use **Edit** to enter counts), and the species management measures with type, owner and status.
- **Attractants within 13 km.** A register of wildlife attractants: landfills, tambak, rice fields, roosts, grazing land and so on. Each entry has a distance, the species it attracts, an action and an owner. Sites inside the 13 km circle are flagged.

The worked example in **DEMO-02** uses the cattle egret, *Bubulcus ibis*. It has 62 strikes in five years, 7 of them damaging, which gives  $5 \times 4 = 20$  (high). It ranks above the barn swallow, which is struck more often but almost never causes damage.



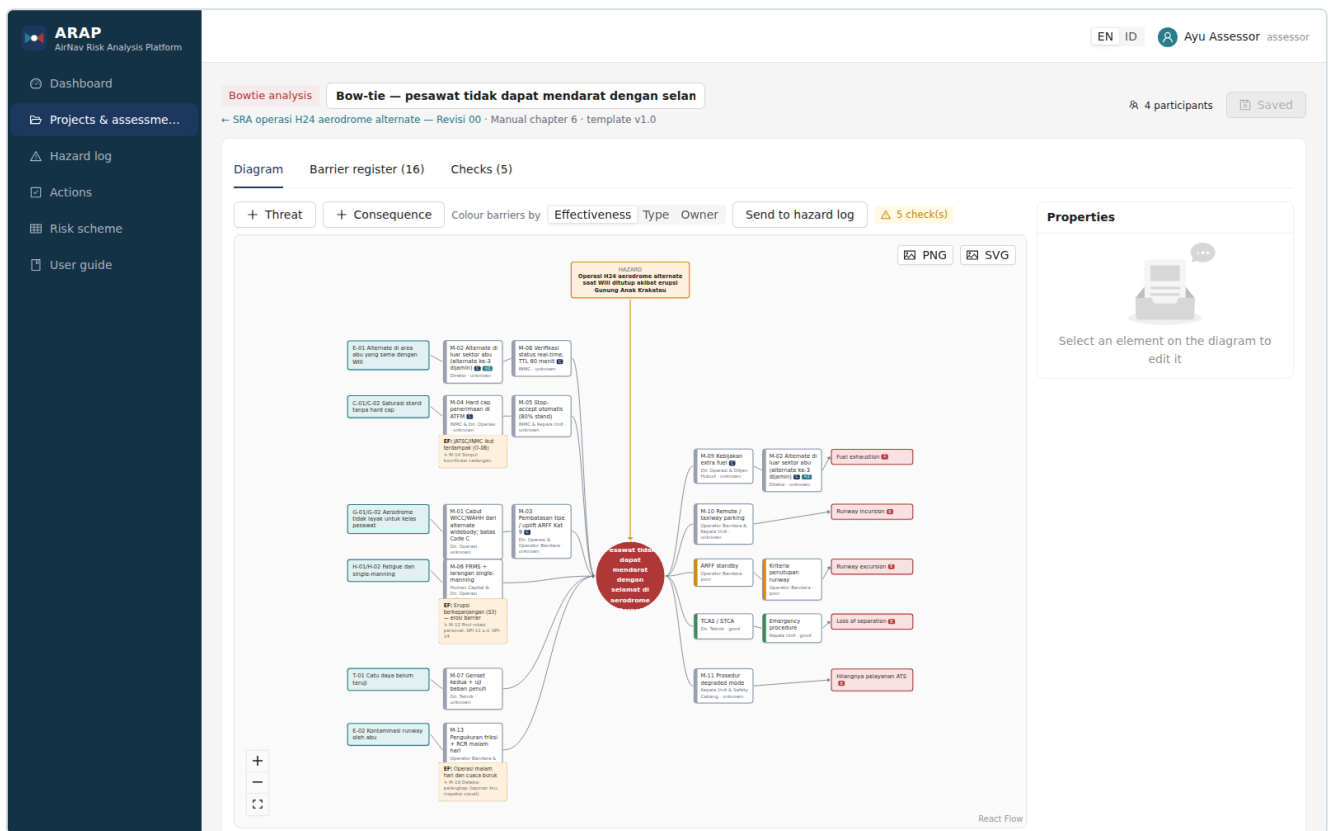
Species profile

### 13.15 Case study: H24 alternate aerodromes (DEMO-03)

The project **DEMO-03** reproduces a real-format Safety Risk Assessment, *SRA/MOC/OPS/001/IX/2026 Rev 00 — Assessment Operasi 24 Jam Aerodrome Alternate*. It covers activating H24 ATS at seven alternates (WAHI, WAHH, WAHQ, WAHS, WICA, WICC, WIPP) when an eruption of Gunung Anak Krakatau closes WIII. The content is kept in Bahasa Indonesia, as in the source report.

| ARAP item                                  | What it holds                                                                                                                                                                                                                      |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Assessment scope, environment, assumptions | The change, system boundary, scenarios S1-S3, demand derivation (72-aircraft design wave), feasibility-gate and capacity-model results, and the assumption register A-01 to A-12                                                   |
| HAZID study <i>Register risiko</i>         | The 40 hazards from FHA (F1-F7), HAZOP and STPA with their current rating, existing barriers and mitigations                                                                                                                       |
| STPA study                                 | The four-organisation closure/reopening decision chain (VAAC → AirNav → operator bandara → KOBU → INMC), six unsafe control actions and their loss scenarios                                                                       |
| Two FTA studies                            | ATS power over 30 days: single untested genset $4.6 \times 10^{-3}$ , two tested gensets $2.9 \times 10^{-4}$                                                                                                                      |
| Bowtie                                     | <i>Pesawat tidak dapat mendarat dengan selamat di aerodrome alternate</i> . New barriers M-xx are marked <i>planned</i> , with the SPI that monitors each one                                                                      |
| GSN                                        | Claim G0 with sub-claims G1-G7; five are <i>undeveloped</i> because they are waiting for preconditions                                                                                                                             |
| Hazard log                                 | ALT-G-01 ... ALT-S-03. The initial rating is the report's <i>Current</i> column, the residual is the <i>Residual</i> column, and the <i>Inherent</i> rating is in the context field. Mitigations M-01 to M-20 are planned controls |
| Actions                                    | ALT-PK-1 ... ALT-PK-8, the activation preconditions (PK-1, PK-3 and PK-4 are blocking)                                                                                                                                             |

The feasibility gate and the capacity (queue/occupancy) model have no dedicated ARAP module yet, so their results are recorded in the assessment's environment text.



DEMO-03 bowtie

## 14. The hazard log

ARAP  
AirNav Risk Analysis Platform

Dashboard

Projects & assessments...

Hazard log

Actions

Risk scheme

User guide

EN ID Ayu Assessor assessor

+ New hazard

↓ CSV

Hazard log

Single register of hazards from every method study, with current and residual risk, controls and owners (Manual §18.2).

Search

Q

Risk region

6 of 6

| Ref     | Hazard                                                   | Unit               | System       | Owner           | Initial | Residual | Controls | Review     | Status |
|---------|----------------------------------------------------------|--------------------|--------------|-----------------|---------|----------|----------|------------|--------|
| HZ-0001 | Unauthorised entry onto active runway                    | Soekarno-Hatta TWR | Aerodrome    | TWR unit chief  | 3A      | 2A       | 2        | 2026-10-18 | open   |
| HZ-0002 | Level bust leading to potential mid-air collision        | Jakarta ACC        | ATM          | ACC unit chief  | 3A      | 1A       | 2        | 2027-01-26 | open   |
| HZ-0003 | Total loss of surveillance picture at ACC sector         | Jakarta ACC        | Surveillance | ACC unit chief  | 3B      | 3B       | 1        | 2026-11-12 | open   |
| HZ-0004 | ATCO fatigue at end of night shift                       | Jakarta ACC        | Roster       | ACC unit chief  | 4C      | 3C       | 1        | 2026-09-23 | open   |
| HZ-0005 | New site outside radio line of sight of low-level sector | Jakarta APP        | VHF          | ACC unit chief  | 3B      | —        | 0        | 2026-11-27 | open   |
| HZ-0006 | Loss of surveillance display at cut-over                 | Jakarta APP        | Surveillance | Project manager | 3C      | 2C       | 1        | 2026-12-27 | open   |

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Hazard log

The hazard log is the single register of hazards from all studies (Manual §32.2). Search, filter by risk region, sort, and export to **CSV**. Overdue review dates are shown in red.

Click a hazard to open it:

ARAP

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Hazard log

Single register of hazards from every method study, with current

Search

Risk region

| Ref     | Hazard                                                   | Unit    |
|---------|----------------------------------------------------------|---------|
| HZ-0001 | Unauthorised entry onto active runway                    | Soeka   |
| HZ-0002 | Level bust leading to potential mid-air collision        | Jakarta |
| HZ-0003 | Total loss of surveillance picture at ACC sector         | Jakarta |
| HZ-0004 | ATCO fatigue at end of night shift                       | Jakarta |
| HZ-0005 | New site outside radio line of sight of low-level sector | Jakarta |
| HZ-0006 | Loss of surveillance display at cut-over                 | Jakarta |

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HZ-0001 — Unauthorised entry onto active runway

Save

Details and risk

Controls (2)

Links

History

\* Hazard

Unauthorised entry onto active runway

Causes / threats

Consequences

Unit

System

Owner

Soekarno-Hatta TWR

Aerodrome

TWR unit chief

Status

Review date

open

2026-10-18

Initial / current risk

Residual risk

clear

|                        | Catastrophic | Hazardous | Major | Minor | Negligible |
|------------------------|--------------|-----------|-------|-------|------------|
| A                      | B            | C         | D     | E     |            |
| Frequent 5             | 5A           | 5B        | 5C    | 5D    | 5E         |
| Occasional 4           | 4A           | 4B        | 4C    | 4D    | 4E         |
| Remote 3               | 3A           | 3B        | 3C    | 3D    | 3E         |
| Improbable 2           | 2A           | 2B        | 2C    | 2D    | 2E         |
| Extremely improbable 1 | 1A           | 1B        | 1C    | 1D    | 1E         |

|                        | Catastrophic | Hazardous | Major | Minor | Negligible |
|------------------------|--------------|-----------|-------|-------|------------|
| A                      | B            | C         | D     | E     |            |
| Frequent 5             | 5A           | 5B        | 5C    | 5D    | 5E         |
| Occasional 4           | 4A           | 4B        | 4C    | 4D    | 4E         |
| Remote 3               | 3A           | 3B        | 3C    | 3D    | 3E         |
| Improbable 2           | 2A           | 2B        | 2C    | 2D    | 2E         |
| Extremely improbable 1 | 1A           | 1B        | 1C    | 1D    | 1E         |

Rationale and evidence for the ratings (required for residual risk)

Occurrence data 2023-2025; barrier audit results

Description / context

Hazard detail

- **Details and risk** — description, causes, consequences, unit, system, owner, status and review date. Rate **initial/current** and **residual** risk by clicking a cell in each matrix. A residual rating requires a **rationale** (the basis of the judgement).
- When you create a new hazard, ARAP checks for **possible duplicates** as you leave the title field.
- **Controls** — each with side (prevention/recovery), type, status, effectiveness, owner and a safety-critical flag. Only *existing-verified* controls may be credited in the current risk.
- **Links** — the source study, the assessment and the related actions.
- **History** — every change from the audit trail.

## 15. Actions

**ARAP**  
AirNav Risk Analysis Platform

EN ID Ayu Assessor assessor

**Actions**  
Risk-reduction actions with owners and due dates. Overdue actions are escalated on the dashboard (COM-12).

+ New action

| Ref      | Action                                                      | Hazard  | Owner           | Due        | Status      |
|----------|-------------------------------------------------------------|---------|-----------------|------------|-------------|
| ACT-0003 | Collect KSS survey for two roster cycles                    | HZ-0004 | FRMS specialist | 2026-09-18 | open        |
| ACT-0004 | Audit stop-bar U/S procedure compliance                     | HZ-0001 | Aerodrome ops   | 2026-09-25 | In progress |
| ACT-0005 | Radio coverage study for new site                           | HZ-0005 | CNS engineering | 2026-10-28 | open        |
| ACT-0001 | Install diverse, physically separate surveillance data path | HZ-0003 | CNS engineering | 2026-12-27 | open        |
| ACT-0002 | Implement SFL mismatch alert in ATM system                  | HZ-0002 | ATM systems     | 2027-02-25 | open        |

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### Actions

Actions have an owner, a due date, a status (open, in progress, closed) and an optional link to a hazard. Overdue actions are flagged here and on the dashboard. **Closing an action requires closure evidence** (e.g. a report number). A Viewer can update an action when the action's owner field is their username.

## 16. Review, approval and locking

| Status                     | Meaning                                   | Next step and who does it                                                                                  |
|----------------------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Draft</b>               | Being prepared; editable.                 | <b>Submit for review</b> — assessor                                                                        |
| <b>In review</b>           | Reviewer checking.                        | <b>Endorse</b> or <b>Return to assessor</b> (comment required) — reviewer                                  |
| <b>Endorsed</b>            | Reviewed; locked.                         | <b>Accept residual risk</b> — authority for the worst residual region; or <b>Reject</b> (comment required) |
| <b>Accepted</b>            | Residual risk accepted; locked.           | <b>Close</b> — reviewer or authority                                                                       |
| <b>Rejected</b>            | Not accepted.                             | Revise and <b>Submit</b> again                                                                             |
| <b>Closed / Superseded</b> | Finished, or replaced by a newer version. | —                                                                                                          |

### Rules ARAP enforces

- An assessment with an **intolerable** residual risk cannot be accepted.
- An assessment with **no hazards** cannot be accepted.
- The person accepting must hold authority for the **worst** residual risk region in the assessment (e.g. *Tolerable (upper)* needs the Accountable Executive).
- Endorsed, accepted and closed assessments are **locked**: studies and hazards become read-only. To change anything, use **New version**. This copies the studies into a new draft (version n+1) and marks the old version *superseded*.

The screenshot shows the ARAP interface for a 'Transition safety assessment' (version 1) which is in an 'accepted' state. The interface includes a sidebar with navigation options: Dashboard, Projects & assessments, Hazard log, Actions, Risk scheme, and User guide. The main content area displays the assessment details, including the scope, environment, and assumptions. The 'Method studies' section shows one study, 'Transition HAZID', with a status of 'open' and an update date of '2026-09-28 10:22'. The 'Hazards' section shows one hazard, 'Loss of surveillance display at cut-over', with an initial risk of '3C', a residual risk of '2C', and one control. The 'Risk acceptance' section shows the worst current/residual region as 'Tolerable (lower)' and the acceptance authority as 'Director of Operations / Engineering'. The 'Actions' section shows zero actions. The 'Review and approval history' section shows two actions: 'endorse by reviewer' and 'accept by director', both dated 2026-09-28 10:23.

Accepted assessment

Every transition is recorded in the **review and approval history** and the audit trail.

## 17. Reports and exports

| What                                    | Where                                   | Format               |
|-----------------------------------------|-----------------------------------------|----------------------|
| Safety Assessment Report (Manual §32.3) | Assessment page → <b>Report (.docx)</b> | Word                 |
| Hazard log                              | Hazard log → <b>CSV</b>                 | CSV (opens in Excel) |
| Bowtie diagram                          | Bowtie editor → <b>PNG / SVG</b>        | Image                |

The report includes scope, methods used, hazards and their risk ratings, study results (fault tree cut sets, LOPA verdict, fatigue indicators, Bayesian network probabilities, and for the extended methods the event-tree outcomes, collision risk, HEPs, availability, simulation statistics, expert weights, occurrence classifications, the GSN argument and the investigation findings), actions, the approval history, and the versions of the risk scheme and calculation engines used.

## 18. Administration

### 18.1 Risk scheme

**Risk scheme** shows the matrix, the tolerability regions with their required action and acceptance authority, and the likelihood levels with their quantitative bands.

The screenshot displays the ARAP (AirNav Risk Analysis Platform) interface. The left sidebar contains navigation links: Dashboard, Projects & assessments, Hazard log, Actions, Risk scheme (selected), and User guide. The main content area is titled 'Risk scheme' and includes a subtitle: 'Active version 1. Changes create a new version and never alter locked assessments (ADM-01).'.

**Risk matrix**

The risk matrix is a 5x5 grid. The columns are labeled 'Catastrophic', 'Major', 'Minor', and 'Negligible'. The rows are labeled 'Frequent', 'Occasional', 'Remote', 'Improbable', and 'Extremely improbable'. The cells are color-coded: red for 'Intolerable', orange for 'Tolerable (upper)', yellow for 'Tolerable (lower)', and green for 'Acceptable'. A tooltip for cell 5B indicates 'Intolerable'.

**Tolerability and acceptance authority**

| Region            | Cells                          | Required action                                            | Authority                            |
|-------------------|--------------------------------|------------------------------------------------------------|--------------------------------------|
| Intolerable       | 3A, 4A, 4B, 5A, 5B, 5C         | Do not introduce or continue; redesign or add controls.    | Cannot be accepted                   |
| Tolerable (upper) | 2A, 3B, 4C, 5D                 | Additional controls; ALARP justification; monitoring plan. | Accountable Executive                |
| Tolerable (lower) | 1A, 2B, 2C, 3C, 3D, 4D, 4E, 5E | Mitigate where reasonably practicable; document ALARP.     | Director of Operations / Engineering |
| Acceptable        | 1B, 1C, 1D, 1E, 2D, 2E, 3E     | Accept; monitor.                                           | Unit / branch manager                |

**Likelihood levels**

| Level | Name                 | Nama                 | Band (per operating hour) |
|-------|----------------------|----------------------|---------------------------|
| 5     | Frequent             | Sering               | 1e-3 and above            |
| 4     | Occasional           | Kadang-kadang        | 1e-5 to 1e-3              |
| 3     | Remote               | Jarang               | 1e-7 to 1e-5              |
| 2     | Improbable           | Tidak mungkin        | 1e-9 to 1e-7              |
| 1     | Extremely improbable | Sangat tidak mungkin | < 1e-9                    |

Risk scheme

Administrators can reassign cells: choose a region under **Paint region**, click the cells, then **Save as new version**. ARAP checks that every cell belongs to exactly one region. Existing assessments keep the version they were created with.

### 18.2 Users

**Users** (administrators only) lists accounts. Click one to edit, or **New user** to add one. Set the role, unit and — for authorities — the risk regions they may accept. Deactivate rather than delete leavers.

### 18.3 Audit trail

**Audit trail** (reviewers and administrators) lists every create, change, approval and export with the user, time and changed fields. Filter by entity.

## 19. Troubleshooting and FAQ

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**I can't edit a study or hazard.** Either your role is Viewer or Authority, or the assessment is endorsed/accepted/closed. Use **New version** on the assessment page (assessor or reviewer).

**"A rationale is required for every risk judgement".** Fill in *Rationale and evidence* in the hazard before saving a residual rating.

**"Residual risk region ... requires ...".** You do not hold authority for the worst residual risk in the assessment. Ask the named authority to accept it.

**The worksheet did not keep my edit.** Finish the edit (Enter or click another cell), then press **Save** in the header.

**My session ended.** Sessions last 8 hours by default; sign in again.

**Numbers differ slightly from the Manual.** The engines reproduce the Manual's worked examples within the tolerances in SRS §9. Rounding in the Manual text is to 2–3 significant figures.

## 20. What version 0.5 does not do yet

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Version 0.5 covers the core workflow and all twenty-six methods. The following SRS items are planned for later releases:

- Single sign-on through Keycloak / SAML / OIDC with MFA. Version 0.2 uses local accounts with JWT tokens.
- Real-time co-editing of a diagram by several users, and the offline workshop pack.
- Imports and exports in BowTieXP XML, Open-PSA MEF, XMLBIF, GeNIe and FRAM Model Visualiser formats.
- File attachments (MinIO), e-mail notifications and reminders.
- Integrations with the occurrence reporting, rostering and maintenance systems.
- CAST (STPA for accident analysis), noisy-OR helpers for Bayesian networks, Monte Carlo uncertainty, and the full “braked” sleep-recovery form of the fatigue model.
- An Apache Superset / Metabase reporting database.
- Linking SPI-register rows to bowtie barriers so barrier health shows the SPI status (SRS SPI-03).
- Saving an existing assessment as a new template, and administrator editing of templates (SRS TPL-05).
- For the wildlife module: import of strike records from the occurrence system and in ICAO IBIS format.
- For the extended methods: import of occurrences from the reporting system into the ERC/RAT module, CRM sensitivity cases, administrator editing of the CARA/HEART libraries, and restricted (need-to-know) access to security studies (SRS SRA-04).